PV and Battery Trading in MGP and MI Auctions Mathematical Formulation, Experiments and Research Extensions

PV–BESS trading research programme

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Abstract

This report documents the implemented decision and execution model for a photovoltaic plant with a shared point of interconnection and a hypothetical AC-coupled battery. It specifies the Italian day-ahead and three intraday auction timeline used by the code, the residual mixed-integer optimization, immutable auction commitments, battery degradation, interval-level feedback controller, optional real-time economic MPC, and ex-post imbalance settlement. It distinguishes the general scenario-tree interface from the deterministic actual-price oracle used in the historical studies. The report follows current source code and shared YAML defaults; it is not a market strategy validation or an investment valuation. It consolidates the main trading implementation, the separate feature and price benchmark work, and the real-time MPC research implementation. Completed forecasting and trading experiments are reported with their sample definitions, information assumptions and numerical limitations. Ongoing searches and implemented but unevaluated extensions are identified separately. The snapshot was taken without launching training, optimization or data acquisition.

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1 Research question and evidence snapshot

The research question is how a PV producer with finite storage should revise auction commitments and physical operation as forecasts and observations arrive. The central idea is to connect three levels: information-aware PV and price forecasts; residual auction optimization with irrevocable submitted quantities; and physical feedback that carries the actual battery state into the next decision. Profit, imbalance, curtailment and battery wear are evaluated on the resulting chronology, rather than inferred from forecast scores or overlapping optimization objectives.

This report audits three distinct working trees. Their implementations have diverged; a feature in one tree is not automatically installed in the others. The main formulation below describes the main trading tree unless explicitly qualified. Section 22 describes the additional MPC research tree. Section 20 describes the forecast studies. The evidence tables refer to saved experiments, which may use earlier policies or models than today's defaults. In particular, old TCN and raw-target results are not relabelled as results of the new reviewed-GHI models.

Working tree	Snapshot HEAD	Contribution
`bess_pv_trading`	`a3238e7`	Auction model, continuous replay, PV studies and reviewed-GHI search.
`bess_pv_trading_features`	`e9d24bd`	Data provenance, DA/IDA1 point and marginal-distribution benchmarks.
`bess_pv_trading_realtime_mpc`	`8645123`	Forecast-economic and adaptive MPC, scenario studies and expanded price searches.

Source hashes, pre-existing working-tree changes and the exact UTC capture time are recorded in `docs/report_evidence/2026-11-04/snapshot.json`. The source LaTeX in all three trees and the probabilistic-PV reference were reconciled with current functions, configuration and saved outputs. Older status prose is superseded only where a completed manifest and numerical evidence exist. "Running" in a saved manifest is a recorded state, not proof of a live process. No annual result is inferred from a short development comparison.

The completed evidence includes an 8,192-candidate reviewed-PV search, the 90-day DA/IDA1 benchmark, January/August auction replays, a 104-job January MPC matrix and a four-farm October MPC week. The newer four-cycle PV search and two expanded price searches remain incomplete at the snapshot. The principal scientific limits are retrospective model selection, assumed publication clocks, synthetic target intervals, unequal ending inventory and short samples.

2 Scope and implementation map

One farm has PV generation, an AC-coupled BESS, and a common AC grid connection. There is no local electricity demand. PV can be exported, stored, or curtailed; the battery can charge from PV or grid imports and discharge toward the grid. The market sequence is MGP (called DA in code), MI-A1, MI-A2, and MI-A3 (IDA1, IDA2, IDA3). Every submitted signed quantity is assumed to execute in full at the auction clearing price. Bid prices, acceptance uncertainty, market impact, and continuous intraday trading are outside this formulation.

The implementation keeps four distinct objects:

1. An optimization plan, which includes the current actionable auction order, future auction recourse, and a battery/PV schedule over the remaining horizon.
2. An optional real-time MPC plan over fixed submitted commitments, with a separately recorded first executable physical action.

3. An immutable commitment ledger, updated at submission and publication.

4. A physical trajectory produced by a feedback/protection policy using realized interval-average PV. Its energy, wear, and deviations are settled separately from the optimization objective.

Source	Responsibility
`src/domain.py`	Validated grids, assets, scenarios, information nodes, states and solver contracts.
`src/optimization/model.py`	Physical, trading, product-resolution, optional imbalance constraints and objective.
`src/optimization/mpc.py`	Common-control real-time MILP and budget-constrained imbalance-price oracle.
`src/optimization/physical.py`	Shared AC battery equations, exclusive controls and degradation.
`src/optimization/pv_envelope.py`	Conditional exact-delivery PV-envelope reduction and reconstruction of original member results.
`src/optimization/degradation.py`	Throughput coefficient and segmented cycle-depth model.
`src/optimization/validation.py`	Numerical validation independent of the solver constraint objects.
`src/transaction_fees.py`	Pure-Python fee integration, dated tariffs and operator monthly state.
`src/optimization/transaction_fees.py`	Flat-band LP costs and exact progressive ABS/PWL constraints.
`src/simulation/synthetic.py`	Shared event replay and auction submission/publication accounting.
`src/simulation/replay.py`	Residual scenario conditioning and bounded physical execution.
`src/simulation/gme_oracle.py`	Manual-data historical price oracle, rolling PV and full-information asset comparisons.
`src/simulation/continuous.py`	Consecutive delivery-day books, shadow boundary projections and uninterrupted execution.
`src/simulation/continuous_inputs.py`	Offline input preflight, deterministic forecast snapshots and optional frozen weather ensembles.
`src/simulation/realtime_control.py`	Interval-start MPC actions and deterministic open-loop shadow controls.
`src/simulation/realtime_mpc_study.py`	Frozen-ledger and continuous comparisons using the same event runner.
`src/reporting/realtime_mpc.py`	Paired controller metrics, plans, signed cash accounting and plots.
`src/reporting/continuous_outputs.py`	Continuous trajectory accounting, paired comparisons and native solve timing.
`src/simulation/settlement.py`	Exact-grid actual settlement and expected/realized KPI reconciliation.
`src/forecasting/models/pv.py`	Weather-to-AC PV conversion.
`src/markets.py`	Configurable local auction gates, publication events and DST-aware delivery grids.

3 Time, units, signs and information

3.1 Delivery intervals and auction products

A local delivery day is partitioned into physical intervals $\mathcal{T} = \{0, \dots, T-1\}$, with starts τ_t, ends τ_{t+1} and durations

$$\Delta_t = (\tau_{t+1} - \tau_t)/\text{one hour}. \tag{1}$$

The historical study uses quarter-hours, so $\Delta_t = 0.25$. A local day is constructed between successive midnights in `Europe/Rome`, then stored in UTC: T is 92, 96, or 100 for spring-DST, ordinary, or autumn-DST days. Repeated local hours are distinguished by their actual UTC offset/fold. No equation assumes that every day has exactly 24 elapsed hours.

A physical interval is not necessarily an independent market product. Before 1 October 2025, DA products are hourly; the MI auctions in the current 2025 history are already quarter-hourly. Physical battery controls are quarter-hourly at all four optimization stages. For the four equal-duration quarters of one native DA hour h, the implementation enforces

$$q_{\mathrm{DA},n,t} = q_{\mathrm{DA},n,t'} \quad (t, t' \in h). \tag{2}$$

These are equal MWh shares of one hourly order, whose total is $\sum_{t \in h} q_{\mathrm{DA},n,t}$. The hourly clearing price is repeated as an objective coefficient on those shares. The quarters do not become separate DA trades or separate observed prices. There is no corresponding date-based hourly equality on battery charging or discharging. Native product resolution and optional physical-control grouping are separate configuration fields.

The configured auction calendar uses local `Europe/Rome` time. The shared calendar validity interval is 1 October 2025 through 31 August 2026; historical study configurations may explicitly override it. The decision cutoff is five minutes before each gate in the supplied studies.

Auction	Gate	Decision cutoff	Results clock	Delivery scope
MGP/DA	D−1 12:00	D−1 11:55	D−1 13:30*	Whole day D
MI-A1/IDA1	D−1 15:00	D−1 14:55	by D−1 15:30	Whole day D
MI-A2/IDA2	D−1 22:00	D−1 21:55	by D−1 22:30	Whole day D
MI-A3/IDA3	D 10:00	D 09:55	by D 10:30	D 12:00–24:00

*The configured 13:30 DA publication is a conservative study assumption; GME's technical rule says final MGP results are not published earlier than 13:05. Recorded publication timestamps override the configured clock.

MI-A1/A2/A3 are the three auction sessions; they exclude the separate continuous MI-XBID market. Every submitted quantity is assumed accepted in full and settled at the auction clearing price. The model does not optimize bid prices, acceptance probabilities, market impact or partial fills. DA products were hourly in the studied data through September 2025. GME made 15-minute MGP products available for delivery from 1 October 2025, and also announced 30-minute products. MI auction products in the studied 2025 data were already quarter-hourly. Product-duration groups, rather than a hard-coded number of variables per day, define trade variables and native-product equalities. The delivery grid is always constructed between local midnights and stored in UTC, so DST days can contain 92, 96 or 100 quarter-hours.

An auction a has an eligibility set $\mathcal{T}_a$. A documented/configured cancellation removes that auction from the active set. Missing data alone does not establish a cancellation: unexplained missing or incomplete curves fail validation. Absent auctions create no trade variables and no submission solve.

At DA closure the integrated model may optimize planned DA, IDA1, IDA2 and IDA3 quantities together. At IDA1, DA is fixed and the remaining three markets may be planned; at IDA2, DA and IDA1 are fixed; at IDA3, the first three are fixed and only IDA3 quantities are free. This integrated planning uses actual future auction curves in the price-oracle experiments, although those prices would not be known at the earlier real-world gates. The sequential baseline removes future-auction trade variables and their expected revenue from each solve. Both modes replan physical operation over the remaining eligible horizon and preserve earlier commitments.

3.2 Notation and electrical basis

Symbol	Unit	Meaning
E	MWh	Nominal internal stored-energy capacity.
$E^{\min}, E^{\max}$	MWh	Operational energy limits derived from SOC percentages.
$\overline{P}$	MW AC	One symmetric battery charging/discharging rating.
$P^{\text{imp}}, P^{\text{exp}}$	MW AC	Directional shared POI ratings.
η_c, η_d	fraction	Complete AC-bus-to-storage and storage-to-AC efficiencies.
σ	fraction/hour	Configured hourly self-discharge fraction.
$V_{s,t}$	MW AC	Forecast available PV in scenario s.
$\widetilde{V}_t$	MW AC	Effective realized PV used for execution.
$c_{n,t}, d_{n,t}$	MW AC	Battery charge/discharge commands at information node n.
$u_{n,t}$	MW AC	PV curtailment.
$i_{s,t}$	MW AC	Uncurtailed PV injection into the common AC bus.
$g_{s,t}$	MW AC	Net grid exchange; positive is export, negative is import.
$e_{s,t}$	MWh	Internal battery energy at an interval boundary.
$q_{a,n,t}$	MWh	Signed auction adjustment; positive sale, negative purchase.
$Q_{s,t}$	MWh	Final cumulative contracted delivery position.
$\pi_{a,s,t}$	EUR/MWh	Auction-price coefficient for interval t.
w_s	fraction	Scenario probability, summing to one.

Power variables are AC at the common bus/POI. Battery energy is internal stored energy. Farm nameplate PV capacity is AC. For pvlib, the DC reference rating is derived as $P_{\text{DC,STC}} = P_{\text{AC,rated}}/\eta_{\text{ref}}$, where the reference factor is a separate, fixed configured assumption (currently 0.96). The physical PVLib path applies the configured DC-to-AC and transformer conversions and caps output at the farm AC rating. Saved neural bundles instead produce an AC prediction subject to their own output bounds and the configured night policy; they do not receive an additional PVLib loss conversion. Measured historical `power` is already AC kW and is converted once to MW; no PV or transformer efficiency is applied to it. BESS efficiency is not applied to measured PV or direct PV exports.

3.3 Forecast vintages and observability

A weather forecast cycle v has source time t_v and availability time A_v. Without explicit source publication times, the configured convention is $A_v = t_v + 6$ hours. At decision cutoff k, the rolling policy chooses the latest eligible cycle with $A_v \leq k$ and required remaining-horizon coverage. Weather cycles about six hours apart are successive forecast vintages, not ensemble members.

The deterministic weather store supplies one forecast member; separate ensemble archives can additionally supply multiple member trajectories.

Measured interval-average PV becomes available after interval end plus configured telemetry delay. A forecast for future intervals is not replaced by future realized power in the ordinary auction replay. If a decision occurs inside an unfinished quarter-hour, the next boundary state is projected through that interval with forecast PV and the same execution policy. Its final realized average cannot yet be observed. The hindsight counterfactuals deliberately have future PV and price information and are labeled as such.

Policy-treated observations carry evidence times. Plateau evidence can require later samples; two-sided interpolation requires its right endpoint and applicable mask evidence. A finalized corrected time series therefore differs from the as-of view available to a past decision. This remains relevant even when a study uses actual-price foresight.

4 PV availability supplied to the optimizer

The optimizer receives nonnegative AC power trajectories $V_{s,t}$ for the represented scenarios. The deterministic farm weather store contains vintages keyed by farm, UTC cycle and hourly lead; successive cycles are not ensemble members. Separate weather-ensemble archives include an explicit member axis. Forecast cycles are selected using publication availability, not just initialization time. The configured availability lag is six hours. Hourly radiation values are aligned to the quarter-hour delivery grid according to their interval-average semantics. PV output is capped at the declared farm AC rating. Historical measured power is already AC and receives no additional conversion loss.

The reference physics path uses configured GHI and horizontal beam irradiance, pvlib solar geometry, fixed-array metadata or the configured single-axis tracker model, plane-of-array irradiance, Faiman module temperature and PVWatts DC output. In symbolic form,

$$G_{\mathrm{DHI}} = G_{\mathrm{GHI}} - G_{\mathrm{BHI}}, \qquad G_{\mathrm{DNI}} \simeq G_{\mathrm{BHI}}/\cos\theta_z, \tag{3}$$

$$G_{\mathrm{POA}} = G_{\mathrm{beam}} + G_{\mathrm{sky}} + G_{\mathrm{ground}}, \qquad T_{\mathrm{mod}} = T_{\mathrm{air}} + \frac{G_{\mathrm{POA}}}{U_0 + U_1 v}, \tag{4}$$

$$P_{\mathrm{DC}} = \max\left(0, \frac{P_{\mathrm{AC,rated}}}{\eta_{\mathrm{ref}}} \frac{G_{\mathrm{POA}}}{1000}[1 + \gamma(T_{\mathrm{mod}} - T_{\mathrm{ref}})]\right). \tag{5}$$

The shared Faiman coefficients are $U_0 = 25$ and $U_1 = 6.84$, and the configured PVWatts coefficient is $\gamma = -0.004/^{\circ}\mathrm{C}$. Historical fit artifacts retain the settings under which they were produced. The DC reference ratio $\eta_{\mathrm{ref}} = 0.96$ is distinct from the configured DC-to-AC scale. The final AC forecast uses the selected output coefficient, transformer setting and AC cap. Farm coordinates are a documented proxy, so the physical geometry is an assumption. The optimization does not use satellite GHI as an operational weather forecast; it is retained for retrospective diagnosis.

For the earlier raw-target modeling studies, the episode-exclusion authority is: `configs/data_quality/pv_excluded_episodes_2025_to_2026_08.csv`. Complete feature/target rows identified there are removed from model splits when the configured exclusion policy is enabled; sequence models do not bridge removed quarters. Those episode exclusions are not automatically applied to the measured PV used by auction execution. The shared `policy.realized_pv_night` instead sets power to zero outside configured month-specific local production windows; the window is applied to each interval's local start, and missing nighttime measurements are also set to zero. The raw store is not modified. `policy.forecast_pv_night` applies the same monthly prior to deterministic and ensemble PV inference, regardless of model. For legacy raw-source runs, missing realized PV is handled using the configured frozen-forecast fallback and missing-interval cap. The reviewed execution profile instead uses reviewed targets, historical-GHI reconstructions

for retained daylight exclusions, and configured night zeros; unresolved daylight gaps fail. Per-interval provenance distinguishes these cases. This optional profile is a counterfactual wherever it uses synthetic production; see Section 20.

4.1 Current registry and historical forecast bundles

The current main-tree replacement registry selects the four reviewed-GHI bundles below. Replacement remains disabled in the shared configuration, so ordinary forecasts still use the configured pvlib path unless a study enables it. The registry availability timestamp is 4 November 2026, 03:09 UTC; historical use requires the explicit retrospective-artifact option.

Farm	Family	Validation-selected configuration
1	Bidirectional GRU	`bigru_extended_021`
345	Transformer	`transformer_extended_001`
901	Bidirectional LSTM	`bilstm_extended_021`
910	TCN	`tcn_extended_003`

Bundles reside under `models/pv/reviewed_ghi_2026_selected/`, with saved weights, fitted preprocessing, farm identity and checksums. Training used historical GHI and preceding-delivery-day 00 UTC weather; validation and testing used forecast GHI. Deterministic inference accepts the latest cutoff-eligible weather cycle, applies the mandatory seasonal night zero and bounds output by AC rating. This later-cycle and changing-horizon use shifts the input/context distribution relative to the completed single-cycle evaluation. It has no completed matched performance claim here. Missing daylight weather cannot be replaced by future observations. A missing/unavailable configured artifact uses the declared deterministic fallback; ensemble preflight requires supported selected artifacts or an explicit pvlib selection.

The historical pilot registry is separate: farm 1 used direct TCN 02, farm 345 TCN 12, farm 901 timing TCN 05 and farm 910 the paired timing TCN. The latter two used lead and cycle-hour channels. Their selection reused inspected 2026 periods. Saved January/August and initial MPC runs retain these models or their explicit farm-345 pvlib override; they were not rerun using the new registry. The mixed-model four-farm comparisons use pvlib with a Genoa location proxy for farm 345 and the retained TCNs for the other farms. Earlier all-TCN comparisons are distinct runs. A city's proxy location is not an identified farm coordinate.

4.2 Optional frozen weather ensemble

When `continuous_optimization.pv_ensemble.enabled` is true, preflight reads the preceding delivery day's 00 UTC ensemble cycle, assumes availability after the configured lag, and requires it to be available at the first cutoff. The cycle is frozen for that delivery day. Each weather member is converted separately to AC PV using the configured PVLib path or a supported weather-bundle model; no ensemble calibration, reweighting or post-processing is implicit. By default, the cutoff-updated deterministic forecast is prepended as another member. If the archive has 50 members, the scenario count is 51 and each weight is 1/51; the optional weather-only sensitivity instead has 50 equal weights. The deterministic member can change at each gate, and at each selected MPC cutoff, while the other 50 paths remain fixed over the day. The weather-bundle inference records its model artifact and retrospective availability; the six-hour historical ensemble availability is an assumption, not a measured publication timestamp. Ensemble bundle inference labels the frozen paths with the DA auction context; the deterministic member is evaluated for the current auction at each gate.

In this PV-scenario mode, auction orders and planned battery commands are shared across members; scenario-specific curtailment provides interval PV recourse, with equal observed PV tied to

the same action. Planning uses the same actual auction price curves across members in the historical oracle studies. The average objective includes any scenario-dependent imbalance terms when they are enabled; in exact-delivery mode shared schedules and prices make the member operating objectives equal. In a scenario formulation, the execution snapshot takes curtailment from the first scenario; the existing grid-tracking controller adapts it to realized PV. Full aligned member values, vintages and input fingerprints are saved separately.

The optional `ensemble_mean` policy uses the pointwise arithmetic mean of the weather members only, including for locked-prefix projections. The optional `ensemble_stochastic` policy preserves the full scenario formulation and replaces exact planned delivery by scenario deviations $\delta_{s,t}$. It maximizes expected operating profit less

$$c \sum_s p_s \sum_t |\delta_{s,t}| + \rho \left(z + \frac{1}{1-\alpha} \sum_s p_s u_s \right), \qquad u_s \geq \sum_t |\delta_{s,t}| - z, \quad u_s, z \geq 0. \tag{6}$$

Here c and ρ are configured EUR/MWh weights, and α is CVaR confidence. The CVaR weight defaults to zero. The stochastic planning imbalance price is zero, rather than a future realized settlement price. The objective uses the configured idealized scenario-dependent curtailment recourse; this is not an optimized real-time controller or an intraday recourse tree. The first scenario is the deterministic member when included; in the weather-only sensitivity it is the first weather member. The unchanged feedback controller executes its planned battery commands and adjusts curtailment against actual PV.

5 Optimization at an auction decision

Each residual solve uses only the remaining suffix of the delivery grid. In the equations below that suffix is reindexed from zero, with T denoting its final boundary; its original UTC product identifiers are retained for commitments.

5.1 Scenario tree and nonanticipativity

At decision k, let $\mathcal{S}_k$ be represented leaf scenarios and let $\mathcal{A}_k$ be the open active auctions. An information tree assigns an auction bid owner $n_a(s)$ to each auction/leaf and a control owner $n_t(s)$ to each interval/leaf. Curtailment may have a separate owner $m_t(s)$ under the selected execution-information convention.

Decisions are node-owned variables, not independent leaf decisions followed by a penalty for disagreement. For example,

$$c_{s,t} = c_{n_t(s),t}, \quad d_{s,t} = d_{n_t(s),t}, \quad q_{a,s,t} = q_{a,n_a(s),t}, \quad u_{s,t} = u_{m_t(s),t}. \tag{7}$$

Thus two scenarios with the same observable history share actionable decisions. A latent PV-member label is not available information by itself. Constraints apply to every represented leaf, including zero-probability leaves if supplied; probabilities only weight the objective. In the current one-scenario historical price oracle, the tree degenerates to deterministic planning with known future prices and a forecast PV trajectory.

5.2 Exact-delivery PV envelope reduction

For a multi-member input with exact interval delivery, common auction orders and battery commands, identical member prices and compatible instantaneous PV-curtailment owners, the optimizer uses one pointwise-minimum PV trajectory. The common position and battery dispatch

imply identical PV injection i_t in every member, so $0 \le i_t \le \min_s V_{s,t}$. Conversely, an envelope solution can be lifted by setting $u_{s,t} = V_{s,t} - i_t$ for each member. Physical costs, trades and objective remain unchanged; zero-probability members still constrain the minimum. The solver reconstructs the original scenario-shaped solution and independently validates it. The full model is retained when these conditions do not hold, including when planned imbalances are enabled. Diagnostics record the original and modeled member counts. This reduction changes neither the weather inputs nor real-time feedback, and equivalence at a fixed planning state does not guarantee identical trajectories when alternative optima are selected. The farm-1 pilot compared the full model and envelope at 37 matched auction states across 2–11 August 2026, then ran each as an independent continuous trajectory under the curtailment-recovery controller. The recorded objectives and realized trajectories agreed to numerical precision on that sample; this is empirical support, not an assertion that independent solvers always choose the same optimum. The study design and limitations are in `docs/pv_lower_envelope_replay.md`.

5.3 Physical variables and balances

For every scenario and remaining interval:

$$i_{s,t} + u_{s,t} = V_{s,t}, \qquad i_{s,t} \ge 0, \qquad 0 \le u_{s,t} \le V_{s,t}, \tag{8}$$

$$g_{s,t} = i_{s,t} + d_{s,t} - c_{s,t}, \tag{9}$$

$$-P^{\text{imp}} \le g_{s,t} \le P^{\text{exp}}. \tag{10}$$

Curtailment appears only in the PV balance. It cannot erase bought grid energy or battery charging energy. A negative grid exchange must be absorbed by net battery charging since there is no local demand. The model does not explicitly assign charging power to its PV/grid source; that partition is an accounting convention used in post-processing.

With $z_{n,t} \in \{0, 1\}$ denoting charging mode,

$$0 \le c_{n,t} \le \overline{P} z_{n,t}, \qquad 0 \le d_{n,t} \le \overline{P}(1 - z_{n,t}). \tag{11}$$

This excludes simultaneous charge and discharge, uses one symmetric AC rating, and introduces no separate battery-inverter power limit.

SOC percentage parameters $s_{\min}, s_{\max}$ define

$$E^{\min} = E\frac{s_{\min}}{100}, \qquad E^{\max} = E\frac{s_{\max}}{100}, \qquad E^{\min} \le e_{s,t} \le E^{\max}. \tag{12}$$

For $\rho_t = (1 - \sigma)^{\Delta_t}$, the transition is

$$e_{s,t+1} = \rho_t e_{s,t} + \eta_c c_{s,t} \Delta_t - \frac{d_{s,t}\Delta_t}{\eta_d}. \tag{13}$$

One AC MWh charged stores η_c MWh. Delivering one AC MWh consumes $1/\eta_d$ internal MWh. In combined mode the two configured efficiencies each cover their entire directional path. In converter-and-cell mode, $\eta_c = \eta_{c,\text{conv}}\eta_{c,\text{cell}}$ and $\eta_d = \eta_{d,\text{conv}}\eta_{d,\text{cell}}$. The alternatives cannot be mixed. The round-trip factor is $\eta_c\eta_d$. Self-discharge acts on stored energy, not on a power variable, and is separate from cycle degradation cost.

The initial boundary is fixed to measured/carried/projected state e^k:

$$e_{s,0} = e^k. \tag{14}$$

If the terminal target is configured, also impose

$$e_{s,T} = e^{\text{target}}. \tag{15}$$

In the GME adapter, initial/terminal percentages are converted using nominal capacity. A `null` terminal SOC omits (15). Prepared and synthetic boundary-state contracts use explicitly named MWh values. Neither version adds a terminal energy salvage value to the objective. An independent optional lower target, configured by `state.terminal_soc_min_percent`, imposes

$$e_{s,T} \geq E\, s_{\text{floor}}/100. \tag{16}$$

Its shared default is `null`. When enabled it applies to every planning horizon end, including intermediate delivery days in continuous operation. A non-null final equality and this floor both apply on the final day. The floor constrains planned energy; forecast-error feedback can produce a realized closing SOC below it while still respecting the hard SOC box.

5.4 Auction quantities and physical delivery

Let Q_t^{fix} contain all previously submitted commitments, priced or still pending. The new final position on leaf s is

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a \in \mathcal{A}_k :\, t \in \mathcal{T}_a} q_{a,n_a(s),t}. \tag{17}$$

An ineligible auction/interval has no trade variable. The model allows signed sales/purchases; they are adjustments to prior positions, not gross generation. For every eligible trade,

$$-P^{\text{imp}}\Delta_t \leq q_{a,n,t} \leq P^{\text{exp}}\Delta_t. \tag{18}$$

For every open auction a in chronological order, define the cumulative planned position after that auction as

$$C_{s,t}^{a} = Q_t^{\text{fix}} + \sum_{b \in \mathcal{A}_k :\, b \leq a,\, t \in \mathcal{T}_b} q_{b,n_b(s),t}. \tag{19}$$

It satisfies

$$-P^{\text{imp}}\Delta_t \leq C_{s,t}^{a} \leq P^{\text{exp}}\Delta_t, \qquad -P^{\text{imp}}\Delta_t \leq Q_{s,t} \leq P^{\text{exp}}\Delta_t. \tag{20}$$

There is no independent commercial volume cap. The directional POI ratings apply to individual auction adjustments, cumulative positions after each auction, and physical delivery. An offsetting later trade is allowed, subject to its own POI-derived limit and the resulting position limit. Thus the model permits bounded cross-auction spread trading even without a battery; its revenues can be present in both asset counterfactuals.

With the shared default `trading.optimize_imbalances: false`, planned delivery is exact in every interval:

$$g_{s,t}\Delta_t = Q_{s,t}. \tag{21}$$

No positive/negative imbalance variables or imbalance direction binary are created. Forecast deviations are not an additional dispatch resource. An optional configured product-group delivery equality can additionally impose $\sum_{t \in h}(g_{s,t}\Delta_t - Q_{s,t}) = 0$. It is redundant with (21) and does not replace it in the disabled-imbalance mode.

5.5 Current objective and sunk cash flows

For throughput degradation with coefficient c_{thr} and an optional import fee f, the current residual objective is

$$\max J_k = \sum_{s \in \mathcal{S}_k} w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t})\Delta_t - f \sum_t [-g_{s,t}]_+ \Delta_t - C_s^{\text{trading}} \right]. \tag{22}$$

The implementation uses an exact import-power linearization with a direction binary only when f is nonzero. The shared default sets this additional import fee to zero; market transaction fees are represented separately below.

5.6 Market transaction fees and operator monthly volume

The `transaction_fees.mode` is `none`, `flat`, or `gme_real` (the shared YAML default). Legacy programmatic inputs without a fee section remain fee-free. Let $m(a)$ be the calendar month of auction submission in market local time, V_m known already executed operator volume, and $G_{m,s} = \sum_{a:m(a)=m,t} |q_{a,n_a(s),t}|$ new gross MWh. Purchases and sales count separately even when final delivery is zero. Quarter-hour energy shares of an hourly product are summed once, without another duration factor. For a cumulative progressive tariff F_m,

$$C_s^{\text{trading}} = \sum_m \left[F_m(V_m + G_{m,s}) - F_m(V_m)\right]. \tag{23}$$

In flat mode this is the configured EUR/MWh coefficient times gross MWh. The published GME MPE marginal rates are 0 on the first 20,000 MWh/month, 0.04 up to 1,000,000, 0.03 up to 10,000,000, and 0.02 thereafter. These bands are progressive and not a retroactive rate on the monthly total.

Fee-free horizons introduce no auxiliary variables. Positive single-band costs use linear absolute-value epigraphs. Across reachable breakpoints, exact Gurobi ABS general constraints and a PWL equality represent both gross volume and the nonconvex tariff; a convex epigraph alone is not used for regressive bands. Absolute-value variables retain auction-node ownership and native-product sharing. Costs are evaluated for each leaf and weighted by w_s.

The fee state includes pending guaranteed transactions immediately at submission. Publication adds no second fee. Executed gross volume remains in V_m during residual solves, while incurred EUR are sunk. The state carries an availability clock and declared provenance; nonzero future executed monthly volume is rejected. Months use transaction dates, so pre-DA/IDA1/IDA2 and IDA3 for one delivery day may belong to different months. A new month starts at zero. The saved final fee state can initialize a later chronological study. Operator scope must explicitly distinguish a standalone plant from a portfolio with exogenous starting volume. Unknown future external activity is not invented: costs are exact conditional on the supplied initial state, not a complete operator-level monthly forecast.

Tariffs have explicit nonoverlapping complete-month validity windows, version, source, thresholds, marginal rates and an applicability status. The current GME tariff and the 2006 annual report show the same structure, the latter effective in 2007. A dated confirmation of 2025 applicability was not established; the shared December 2024–December 2026 tariff marks this explicitly as `assumed_unchanged`. Unsupported dates fail. The shared initial state assumes a standalone simulated operator at local midnight on 1 December 2024, with zero December volume, zero earlier fees and no external portfolio activity. Subsequent months begin at zero and continuous execution carries actual submitted volume. Known operator-level volume should replace this assumption when the plant belongs to a larger portfolio. Enabling this tariff can still produce zero variable fees below the monthly exemption. Example YAMLs are `configs/full_experiments/synthetic_fees_none.yaml`, `configs/full_experiments/synthetic_fees_flat.yaml`, and `configs/full_experiments/synthetic_fees_gme.yaml`. Membership, taxes, ACER record charges and separate aggregator commissions remain excluded; physical grid fees and degradation remain separate cost components.

Past auction cash flow and already incurred degradation are sunk at residual reoptimization. Priced commitments remain in the external ledger, and pending unpriced submitted quantities enter Q^{fix} without adding an optimizable cash-flow term. Their future clearing value is constant with respect

to all remaining decisions and is excluded from J_k. This avoids double counting. The initial pre-DA solve has no prior commitments; its objective spans the modeled whole-day opportunity set. Residual objective values therefore should not be summed across auction solves to compute daily profit.

There is no explicit curtailment penalty, risk measure, capital cost, fixed O&M, tax, brokerage fee, ramp constraint, minimum dwell time, or calendar-aging charge in (22). Curtailment can affect revenue through delivery choices, but is not independently priced. The horizon can exploit inventory when its terminal target is free; this matters when comparing asset value.

5.7 What is free at each of the four decisions

Decision	New free auction quantities	Previously submitted fixed quantities
DA	DA, IDA1, IDA2, IDA3 plans	None
IDA1	IDA1, IDA2, IDA3 plans	DA
IDA2	IDA2, IDA3 plans	DA and IDA1
IDA3	IDA3 eligible products	DA, IDA1 and IDA2

At each decision, remaining-horizon charge/discharge modes, curtailment, grid powers and energy states are free subject to their balances, boundary state, nonanticipativity, product rules and terminal condition. Executed past intervals are absent from the residual grid. Future-auction quantities are plans until their own submission. Only the current auction order is committed from that solve. Before noon, IDA3 cannot change orders for the morning; physical control commands from the previous plan also remain in force. All IDA3 decision variables cover only the 48 quarter-hours from 12:00 to midnight: charge, discharge, mode, curtailment, grid power and trades, with 49 SOC boundaries. Gate closure determines the information cutoff, not the delivery horizon.

More generally, let c_k be the submission cutoff and j_k the first grid boundary no earlier than both c_k and the current auction's first eligible delivery. The solver starts at j_k. From the latest observed physical state, the replay projects the locked prefix using the previously requested commands $\bar{u}_i$, fixed contracted grid targets Q_i/Δ_i, and the latest cutoff-available PV forecast $\widehat{P}_i^{(c_k)}$ through the same physical controller Φ used for delivery:

$$\widehat{x}_{i+1|c_k} = \Phi\left(\widehat{x}_{i|c_k}, \bar{u}_i, Q_i/\Delta_i, \widehat{P}_i^{(c_k)}\right), \qquad i < j_k. \tag{24}$$

Here x includes energy and degradation state. The resulting $\widehat{E}_{j_k|c_k}$ is the fixed initial energy of the residual MILP. Forecast publication eligibility is still checked at c_k; projecting to noon does not permit a forecast released after the 10:00 gate. No future measured PV is used in this projection. Actual delivery continues with the retained commands and realized PV, so its noon state may differ from the projected state. Grid feedback and protection enforce the actual SOC and power bounds. If IDA3 is the first participating auction, zero-position feedback continues until noon.

6 Battery degradation models

6.1 Current AC throughput approximation

For one trajectory,

$$H = \sum_t (c_t + d_t)\Delta_t, \qquad D_{\text{thr}} = c_{\text{thr}} H. \tag{25}$$

Both directional AC throughputs are charged once. The YAML can directly supply c_{thr} or derive it from replacement economics and reference cycle life. Let p_{USD} be pack USD/kWh, x be USD

per EUR, p_{extra} additional EUR/kWh replacement cost, s_R salvage fraction, N reference cycle life and d_{ref} reference internal depth of discharge. Then

$$R = 1000E\left(\frac{p_{\text{USD}}}{x} + p_{\text{extra}}\right)(1 - s_R), \tag{26}$$

$$H_{\text{life}} = Nd_{\text{ref}}E\left(\frac{1}{\eta_c} + \eta_d\right), \tag{27}$$

$$c_{\text{thr}} = \frac{R}{H_{\text{life}}}. \tag{28}$$

One reference internal cycle charges $d_{\text{ref}}E/\eta_c$ AC MWh and discharges $d_{\text{ref}}E\eta_d$ AC MWh, explaining the normalization. There is no additional factor of two or half applied to (28). The configured end-of-life capacity fraction is descriptive metadata, not an extra multiplier or a dynamic capacity update.

The shared YAML uses a historical generic pack-price proxy, a dated FX conversion, and an illustrative reference cycle life. Those are study inputs with source metadata, not an estimate of a specific installed BESS cost. The resolved R and c_{thr} are saved. The cost is financial wear; it does not reduce E, $\overline{P}$, or efficiencies as the simulation proceeds.

6.2 Optional segmented depth-dependent aging

The optional `xu_segmented` mode follows the configured segmented cycle-depth construction described in the repository's Xu notes and the author paper [1]. It is distinct from the current throughput study. For a configurable convex stress curve $\Phi(d) = \alpha d^{\beta}$, $\beta \geq 1$, partition nominal internal capacity into J equal segments of size E/J. The marginal AC discharge cost of segment $j = 1, \ldots, J$ is

$$m_j = \frac{RJ}{\eta_d E}\left[\Phi(j/J) - \Phi((j-1)/J)\right]. \tag{29}$$

It is nondecreasing with depth. The degradation charge is

$$D_{\text{Xu}} = \sum_t \sum_{j=1}^{J} m_j d_{j,t} \Delta_t. \tag{30}$$

Only discharging is charged under this convention. A discharging half-cycle incurs the convention's depth cost, while a charging half-cycle does not; do not add an extra factor of one half. Nonzero throughput cost is rejected in this mode.

Segment flows and energies obey

$$\sum_j c_{j,t} = c_t, \qquad \sum_j d_{j,t} = d_t, \qquad \sum_j h_{j,t} = e_t, \tag{31}$$

$$h_{j,t+1} = h_{j,t} + \Delta_t(\eta_c c_{j,t} - d_{j,t}/\eta_d), \qquad 0 \leq h_{j,t} \leq E/J. \tag{32}$$

Segment energies represent cycle-depth memory, not fixed absolute SOC buckets. Charging and discharging both use shallow available segments first. For each $j < J$, the implementation adds priorities $v^c_{j,t}, v^d_{j,t} \in \{0, 1\}$:

$$\sum_{\ell > j} c_{\ell,t} \leq \overline{P} v^c_{j,t}, \qquad h_{j,t+1} \geq (E/J) v^c_{j,t}, \tag{33}$$

$$\sum_{\ell > j} d_{\ell,t} \leq \overline{P} v^d_{j,t}, \qquad h_{j,t+1} \leq (E/J)(1 - v^d_{j,t}). \tag{34}$$

Using a deeper segment therefore requires the shallower segment to end full when charging or empty when discharging. This chooses a deterministic shallow- first representative, including otherwise

indifferent charging allocations. Segment actions have the same information ownership as aggregate battery power.

At a declared fresh study start, initialize

$$h_{j,0} = \operatorname{clip}\big(e_0 - (j-1)E/J,\ 0,\ E/J\big). \tag{35}$$

Afterwards the segment vector must be carried with actual execution; an arbitrary residual solve cannot reconstruct it from aggregate SOC. Previous throughput or aging requires explicit memory. Xu mode currently rejects nonzero self-discharge. It remains a finite-segment approximation and omits calendar, temperature and current-rate aging, capacity-fade feedback and unknown pre-study cycle history.

7 Submission, publication and reoptimization workflow

Before each active auction submission, the GME replay refreshes the latest available PV forecast over unexecuted intervals, projects a boundary if necessary, and solves the residual planning problem. Previously submitted quantities, including still-unpriced ones, are fixed. At submission, only the current auction's node-shared action is extracted. Later planned auction quantities are not booked and can change.

At publication, known prices are updated, and actual contractual revenue $\pi^{\text{real}}_{a,t} q^{\text{sub}}_{a,t}$ is entered exactly once in the ledger. Publication does not reopen a market. The current GME experiment forces `reoptimize_physical_controls=false`; it uses four auction solves for four active auctions, not an extra set of publication solves or physical optimizations. Physical feedback is algebraic under this retained-plan policy. The optional continuous MPC policy adds interval-start solves as described below. Synthetic replay can separately enable interval-control reoptimization, so its solve count need not be four.

At the next auction, closed commitments are carried, remaining open opportunities are reconsidered, and past executed state/aging remain fixed. In the actual-price oracle all future auction prices are already supplied as a benchmark, so price publication does not remove price uncertainty there. In a forecast-based route, known results condition subsequent price distributions without changing the commitments that produced them.

8 Optional real-time economic MPC

The default remains `track_committed_grid`. Four additional continuous execution selectors combine deterministic/ensemble PV with price-blind/oracle imbalance treatment: `deterministic_mpc`, `deterministic_mpc_price_oracle`, `ensemble_mpc`, and `ensemble_mpc_price_oracle`. They add physical control solves at each quarter-hour start, over the remainder of that local delivery day. Auction submission, publication, fees and eligibility are unchanged. In particular, planned future orders are not fixed positions in this lower-level model.

For each MPC horizon, Q^{sub}_t contains only quantities already submitted, including unpriced commitments. Battery powers c_t, d_t, mode binaries, energy e_t and a utilization fraction $a_t \in [0,1]$ are common across *all* scenarios for the *entire* horizon. The additional equations are

$$i_{s,t} = a_t V_{s,t}, \qquad g_{s,t} = i_{s,t} + d_t - c_t, \tag{36}$$

$$\delta_{s,t} = g_{s,t}\Delta_t - Q^{\text{sub}}_t, \qquad L = \sum_s w_s \sum_t |\delta_{s,t}|. \tag{37}$$

The shared battery transition (13), power exclusivity, efficiencies, SOC box and POI bounds apply. MPC and auction builders reuse common battery and degradation helpers. Scenario-specific battery

recourse and arbitrary scenario-specific curtailment are absent. The first utilization fraction is applied to realized available PV by the execution environment after issuance.

Price-blind MPC minimizes

$$\kappa L + D^{\text{plan}} + C^{\text{import}} + C^{\text{terminal}}, \qquad \kappa = 500 \text{ EUR/MWh by default.} \tag{38}$$

There is no independent curtailment penalty. The default terminal cost is $50|e_T - 0.5E|$ EUR, with energy in MWh; no-value and configurable linear-value alternatives are supported. Separately configured final-study hard targets and explicit horizon-end minima remain constraints. A hard 50% equality is not imposed on ordinary operating midnights. Planning terminal costs never become real cash charges.

The imbalance-price oracle first minimizes L with the same physical and hard terminal constraints. After solver certification within the recorded tolerance, it maximizes expected *signed* settlement minus degradation, import and terminal costs, subject to

$$L \le L_{\min}(1 + r) + b. \tag{39}$$

The relative and absolute budgets r, b default to zero. Actual future Terna prices enter only an explicitly marked noncausal input object; price-blind MPC does not accept it. Single-price cash flow is $\lambda_t \delta_{s,t}$, including negative prices, rather than $-|\lambda_t||\delta_{s,t}|$. Distinct directional curves retain sign exclusivity. With a single curve, the nonnegative split is exact in the minimum-volume stage and conservatively bounds true absolute volume in the economic stage; reporting reconstructs physical volume directly.

One configurable ten-second wall budget covers building and both solver stages. Stage objectives, bounds, gaps and actual additional imbalance are logged. An uncertified first-stage incumbent cannot authorize an economic second stage; it can supply a validated price-blind action. If economic refinement has no usable solution, a validated first-stage action is retained. No valid action invokes the explicitly logged existing tracking fallback. Solver tolerances remain relative MIP gap 10^{-4} with no configured absolute gap.

Information comprises current observed energy and degradation memory, completed interval PV, submitted commitments, issued commands and eligible forecast vintages. By default current and future realized PV are absent from MPC inputs. The explicit `realtime_mpc.current_pv_information` option can instead disclose the realized current-interval average before solving. It replaces $V_{s,0}$ by that same measured/effective PV value in every member while all later intervals remain forecast-only. This is a retrospective current-PV oracle, separate from the imbalance-price oracle; both disclosures are recorded. The ensemble uses the preceding delivery day's frozen 00 UTC weather members and a latest eligible deterministic member, uniformly weighted by default; explicit normalized weights and a weather-only sensitivity are supported. The weights are not calibrated probability claims. The revised two-day Farm 1 examples use the selected `direct_tcn_02` bundle for deterministic/ensemble conversion, initialization, measurement fallbacks and the common auction-source forecasts, without fitting. Its training period is 2025; its January–April 2026 selection and October 2026 artifact registration make January reuse retrospective. Checksums and this hindsight selection are retained. The initial pvlib/current- forecast comparison is preserved separately; the operational default remains pvlib with replacement disabled.

Only the first action executes, through `follow_battery_plan` protection without economic grid-tracking replacement. Interval-average realized PV can change controls only through declared physical protection. Full plans, first commands and overrides are saved separately. Auction events cannot retrospectively replace an unfinished interval's command.

MPC auction-boundary projection follows the last completed open-loop plan from the latest actual state, preserving an outstanding action and using the latest eligible *deterministic* PV forecast at the auction cutoff, even for ensemble MPC. It does not solve hypothetical future MPCs or choose

a weather member. Missing plan coverage invokes logged forecast-only tracking/protection. In current-PV-known mode the locked unfinished interval uses the scalar already disclosed at issuance, while subsequent intervals retain deterministic forecast PV; the shadow rollout never retrieves future actual production. The result, `projected_soc_mpc_deterministic`, never overwrites actual state. Pure battery-power commands can make projected energy independent of PV until protection acts. Realized noon/midnight projection errors are recorded; actual carryover is not reset to tomorrow's earlier assumed state.

The original two-day diagnostic freezes one existing integrated deterministic auction ledger and replays identical timed submissions with each controller. Auction revenue and fees must match. The full continuous comparison lets actual SOC and MPC projections change later auction orders. Their interpretations remain separate. The auction-price oracle is unchanged and distinct from the new lower-level imbalance-price oracle. Tracking's interval-average PV feedback advantage is disclosed in causal mode; the current-PV-known variant supplies the same interval-average production to MPC. An optional causal tracking diagnostic is available. Detailed configuration, metrics, limitations and commands for this main-tree implementation are in `docs/realtime_economic_mpc.md`. The newer sibling research launcher runs continuous comparisons as its primary design, with additional economic/adaptive controllers and revised oracle budgets described in Section 22.

9 Physical feedback with realized PV

9.1 Feasible interval power range

The current policy is `track_committed_grid`. For interval t its target is

$$g_t^\star = Q_t^{\text{committed}}/\Delta_t. \tag{40}$$

The controller uses effective realized interval-average PV $\widetilde{V}_t$; this is an idealized interval simulator, not a subsecond telemetry/control model. It does not disclose this final average to the earlier auction optimizer. Given actual initial energy e_t and retained energy $\bar{e}_t = \rho_t e_t$, the feasible charging/discharging maxima are

$$C_t = \min\left\{\overline{P}, \frac{[E^{\max} - \bar{e}_t]_+}{\eta_c \Delta_t}\right\}, \tag{41}$$

$$D_t = \min\left\{\overline{P}, \frac{\eta_d \left[\bar{e}_t - E^{\min}\right]_+}{\Delta_t}\right\}. \tag{42}$$

First clip the planned charge/discharge commands to these maxima, obtaining c_t^0, d_t^0. Simultaneous positive requested charging and discharging after protection is rejected. Initially curtail only available PV: $u_t^0 = \min\{\widetilde{V}_t, \left[u_t^{\text{plan}}\right]_+\}$.

9.2 Grid-tracking policy

Release scheduled curtailment if it can help meet a deficit:

$$g_t^0 = \widetilde{V}_t - u_t^0 + d_t^0 - c_t^0, \tag{43}$$

$$\bar{u}_t = u_t^0 - \min\{u_t^0, \left[g_t^\star - g_t^0\right]_+\}. \tag{44}$$

The desired net battery dispatch and its feasible projection are

$$b_t^{\text{req}} = g_t^\star - \widetilde{V}_t + \bar{u}_t, \tag{45}$$

$$\tilde{b}_t = \text{clip}(b_t^{\text{req}}, -C_t, D_t). \tag{46}$$

Finally curtail any remaining surplus that battery adjustment cannot absorb:

$$\tilde{u}_t = \bar{u}_t + \min\{\widetilde{V}_t - \bar{u}_t, \left[\widetilde{V}_t - \bar{u}_t + \tilde{b}_t - g_t^\star\right]_+\}. \tag{47}$$

The controller then recovers curtailed PV using the available battery headroom, without changing grid exchange. It reduces discharge before increasing charge:

$$r_t = \min\{\tilde{u}_t, \left[\tilde{b}_t + C_t\right]_+\}, \tag{48}$$

$$b_t = \tilde{b}_t - r_t, \qquad u_t = \tilde{u}_t - r_t, \tag{49}$$

$$c_t = [-b_t]_+, \qquad d_t = [b_t]_+. \tag{50}$$

This is an unconditional feedback action, rather than a new optimization; its SOC and degradation effects carry forward to later intervals. The saved `curtailment_recovered_mw` measures the curtailment reduced in this feedback step, whose starting value depends on the active plan. It can include avoided discharge as well as extra charging and depends on the plan; it is not itself a direct measure of energy newly charged into the battery. Since the first-member plan is used by the controller, reconstruction of that member's curtailment after envelope reduction is necessary for consistent execution. Thus excess PV first decreases discharge, then increases charging as needed and feasible, then is curtailed. A PV shortfall first releases planned curtailment, then reduces charging or increases discharge within the actual SOC box. The remaining tracking error is preserved if battery power/energy cannot supply it.

POI protection follows. If net import exceeds its rating, charging is reduced. If export exceeds its rating, discharge is reduced first and remaining excess PV is curtailed. Grid power is recomputed consistently with the adjusted actions. Actual energy then advances using (13) with these executed powers, and bounds are checked before accepting the new state. Both throughput/Xu memory and cumulative financial wear are advanced from executed actions, not from the old auction plan. Protection is mandatory under either execution policy; the alternative `follow_battery_plan` skips grid-tracking adjustment.

There is no controller constraint forcing the actual terminal energy to the auction plan's target. Forecast errors and grid tracking can cause an ending-SOC difference even if every optimization imposed (15). This is visible in the current terminal-50% study and is not equivalent to guaranteeing 50% realized SOC. Any hard actual-terminal policy would be an additional controller design and could create different imbalances/cash flows.

9.3 PV source allocation for reporting

For the executed trajectory, use the explicit PV-first accounting convention

$$P_t^{\mathrm{PV}\to\mathrm{B}} = \min\{\widetilde{V}_t - u_t, c_t\}, \tag{51}$$

$$P_t^{\mathrm{PV}\to\mathrm{G}} = \widetilde{V}_t - u_t - P_t^{\mathrm{PV}\to\mathrm{B}}, \tag{52}$$

$$P_t^{\mathrm{PV}\to\mathrm{curt}} = u_t. \tag{53}$$

They sum to $\widetilde{V}_t$ in every interval. Grid-supplied charging and battery discharge are not counted as PV production. These are post-processing flows; there are no separate source-attribution decision variables in the MILP.

10 Settlement and expected-versus-realized accounting

Actual net deviation is retained at the native quarter-hour:

$$\delta_t^{\mathrm{real}} = g_t^{\mathrm{real}}\Delta_t - Q_t^{\mathrm{committed}}. \tag{54}$$

Positive means more net export than promised. With the selected single Terna NORD settlement price λ_t, the modeled cash flow is

$$S_t = \lambda_t \delta_t^{\text{real}}. \tag{55}$$

A positive price therefore does not mean every imbalance is penalized: positive excess export is credited and a negative deviation is debited. Negative prices are allowed and reverse these cash-flow signs. The general settlement helper also accepts two directional curves:

$$S_t = \lambda_t^+ \left[\delta_t^{\text{real}}\right]_+ - \lambda_t^- \left[-\delta_t^{\text{real}}\right]_+. \tag{56}$$

This documents the implemented study accounting, not every possible Italian participant-specific settlement item or regulatory regime. The workbook adapter selects the declared NORD macrozone and the imbalance-price column, retaining other imported fields as provenance. Product starts and ends must match the execution grid exactly. Prices are neither hourly-averaged nor filled across gaps; missing enabled settlement prices are an error.

Actual daily auction cash flow and settled operating profit are

$$A^{\text{real}} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{\text{sub}}, \tag{57}$$

$$J^{\text{real}} = A^{\text{real}} + \sum_t S_t - D^{\text{executed}} - f \sum_t \left[-g_t^{\text{real}}\right]_+ \Delta_t - C^{\text{trading,real}}. \tag{58}$$

The stochastic-ensemble model deterrence penalty is not an actual settlement charge and is not subtracted from realized profit. Legacy non-ensemble runs with optimized imbalance retain their existing configured realized penalty. If settlement is unavailable, profit before settlement is reported and fully settled profit remains unknown in the settlement helper and compatible independent-day outputs. The continuous comparison runner requires complete actual settlement prices and rejects a run when settlement is disabled or incomplete. The continuous ledger additionally decomposes realized auction cash flow as

$$A^{\text{real}} = \sum_t \pi_{DA,t}^{\text{real}} \sum_a q_{a,t}^{\text{sub}} + \sum_{a \neq DA,t} (\pi_{a,t}^{\text{real}} - \pi_{DA,t}^{\text{real}}) q_{a,t}^{\text{sub}}. \tag{59}$$

Each signed quarter-hour quantity is in MWh, including the allocated shares of native hourly products; no second duration factor is applied. If the DA reference curve is absent, both components are unavailable rather than filled. Offsetting volume is the sum across products of the smaller of gross purchases and gross sales. Matched PV-only continuous replays remove battery capacity and power while retaining the forecast and market policy. Their cash-profit difference does not assign a terminal value to remaining stored energy. Only actually submitted guaranteed transactions enter $C^{\text{trading,real}}$; future plans that are never submitted contribute no realized fees. The ledger retains transaction-month volume, per-trade charges and tariff versions. An independent Python calculation reconciles those charges and the final monthly state. Expected and realized transaction fees are reported separately from gross auction revenue. In `realized_profit_breakdown.png`, trading-fee EUR and gross MWh occupy the former import-fee slot; nonzero physical grid fees appear as their own additional bar.

Let J^{init} and A^{init} be the initial auction solve's expected profit and auction revenue. Saved mismatches are

$$\Delta J = J^{\text{real}} - J^{\text{init}}, \qquad \Delta A = A^{\text{real}} - A^{\text{init}}, \qquad r_J = 100 \frac{\Delta J}{|J^{\text{init}}|}. \tag{60}$$

The percentage is undefined/`null` when its expectation is zero; unsettled profit comparisons remain unknown. Differences include later replanning and physical execution effects, not only price-forecast error.

The replay also evaluates each stored auction snapshot as an open-loop diagnostic against realized PV and the same settlement prices. Its planned battery commands are protected physically. Deviations are measured against that snapshot's planned grid schedule (which can include assumed future trades) or the final contract, as labeled in the recorded comparisons. It is not another auction optimization. Because IDA3 covers a shorter residual horizon, a common-horizon comparison is also saved; unaligned whole-horizon totals should not be compared as if their delivery intervals were identical.

11 Value of reoptimization through frozen-plan replay

The optional frozen-plan comparison asks what would happen if planning stopped after one auction decision but physical feedback continued. This differs from the older open-loop imbalance diagnostic, which follows planned battery commands without grid-tracking feedback and does not reconstruct an alternative complete auction ledger and net profit.

Let k be an active auction other than the final active auction. At its cutoff, retain the entire deterministic future-trade vector from that solve. Define

$$q_{a,t}^{[k]} = \begin{cases} q_{a,t}^{\text{submitted,ref}}, & a \le k, \\ q_{a,t}^{\text{plan},k}, & a > k, \end{cases} \qquad Q_t^{[k]} = \sum_a q_{a,t}^{[k]}. \tag{61}$$

The already submitted trades are fixed even if their prices are unpublished. Future orders are the frozen snapshot's orders, not the later realized orders from the reoptimized strategy. They are submitted at their original gates and priced at actual clearing prices. Their eligibility must ensure submission precedes the relevant delivery interval. Thus their cumulative position is a valid delivery-time tracking target without prematurely revealing new prices or forecasts to the frozen plan.

Executed history before the frozen residual boundary remains identical to the reference trajectory. The suffix starts from the actual carried energy, throughput, incurred wear and, for Xu mode, segment-memory vector. It is not reset to the snapshot's forecast SOC, including when the cutoff projects a not-yet-observed boundary. At physical delivery that boundary's actual state is carried forward without changing the frozen decisions. The same configured execution policy uses the frozen battery/curtailment commands, target $Q_t^{[k]}/\Delta_t$, and the same effective realized PV:

$$(c_t^{[k]}, d_t^{[k]}, u_t^{[k]}, e_{t+1}^{[k]}) = \mathcal{F}(e_t^{[k]}, \widetilde{V}_t, Q_t^{[k]}/\Delta_t, c_t^{\text{plan,k}}, d_t^{\text{plan,k}}, u_t^{\text{plan,k}}). \tag{62}$$

Here $\mathcal{F}$ is the grid-feedback/protection policy described above. The current study uses grid tracking; the alternative execution policy, if selected, is shared with the reference. No later optimization or weather-forecast update is applied. The algebraic replay adds zero optimization solves.

Using the same signed settlement convention and actual dated transaction-fee accounting as the reference, the full-day frozen profit is

$$J_{\text{frozen}}^{[k]} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{[k]} + \sum_t S_t^{[k]} - D^{[k]} - C_{\text{import}}^{[k]} - C_{\text{trading}}^{[k]}. \tag{63}$$

Previously executed prefix costs are preserved. Future wear is recomputed from the counterfactual's executed commands; gross fees are recomputed from its own trade ledger and the same initial operator-volume state, including month transitions and tariff bands. Reference future trades or fees are never borrowed after the freeze point.

The value KPI is

$$G_k = J_{\text{fully reoptimized}}^{\text{real}} - J_{\text{frozen}}^{[k]}. \tag{64}$$

Both sides cover the entire delivery day, so differing residual lengths do not create incomparable profit totals. Positive values favor later reoptimization; negative values are retained. Each G_k measures all later replanning after k: the values are not additive and need not be monotonic. This comparison includes new forecasts, revised orders and state-aware planning rather than isolating a causal effect of forecast accuracy. Final stored-energy differences are reported; no terminal salvage price is assumed. If actual settlement is unavailable, settled profit and G_k remain unknown.

The independent YAML section is `reoptimization_baseline`. Its options are `enabled`, `freeze_after_auctions`, and `plots`. An absent section disables the feature; the comparison study enables it. A null auction selection evaluates every active pre-final auction, while a list chooses a subset. The last active auction is always excluded, cancelled auctions produce no baseline, and an empty selection produces no comparisons. The plot switch also respects the global plot setting. Snapshot replay requires deterministic plans; it rejects using one selected leaf of a contingent plan as if all of its future branches were already known.

The simulator is `src/simulation/frozen_plans.py`. Saved dispatch-plan rows include every `planned_trade_<auction>_mwh` and their planned net position. Full-day profits, gains and states are saved in the frozen-plan summary CSV/JSON, with separate actual counterfactual dispatch and trade ledgers. Individual gains also enter the scalar KPI table, and `reoptimization_value.png` compares frozen profits with the reference. The isolated January configuration is `configs/full_experiments/gme_oracle_2025-01-01_farm910_reoptimization_value.yaml`.

12 Perfect-information schedules and the PV-only baseline

12.1 PV+BESS versus the feedback controller

After replay, one additional full-day problem uses $V_t = \widetilde{V}_t$ and all actual auction-price curves, with a single scenario and all auction orders jointly free subject to their native products, eligibility, POI and physical constraints. It retains the BESS ratings, efficiencies, SOC box, degradation, initial energy and configured terminal target. Call its achieved solver objective $J^{\mathrm{PI}}_{\mathrm{PV+BESS}}$.

This is deliberately a hindsight plan rather than a causally executable policy. It is compared with the feedback controller's actual dispatch, curtailment and SOC, and its objective is saved as the solver's `ObjVal`. Four missing measurement intervals, if permitted by the selected run, are handled through that run's explicit fallback; diagnostics record their count and identities. The schedule figure shows both trajectories and their battery-power difference through the final day boundary. It does not assert that feedback should reproduce the hindsight trajectory or terminal energy.

12.2 PV-only asset counterfactual

When `experiment_outputs.pv_only_baseline: true`, a second independent solve uses the same optimizer builder with

$$E = 0, \quad \overline{P} = 0, \quad E^{\min} = E^{\max} = 0, \quad e_0 = e_T = 0, \quad D = 0. \tag{65}$$

Battery powers and energy variables are consequently fixed to zero; the code reuses one physical/trading formulation instead of implementing a second, potentially inconsistent PV-only model. Unused variables can be removed by solver presolve. PV injection/curtailment and auction trades remain free subject to the same effective realized PV, native products, active auctions and POI. Call its achieved objective $J^{\mathrm{PI}}_{\mathrm{PV}}$.

The incremental operating-value KPI is

$$\Delta J^{\mathrm{PI}}_{\mathrm{BESS}} = J^{\mathrm{PI}}_{\mathrm{PV+BESS}} - J^{\mathrm{PI}}_{\mathrm{PV}}. \tag{66}$$

The BESS side includes its own modeled degradation; the PV-only side has no battery wear. Both sides retain permitted cross-auction spread trading, so market-trading revenue common to both is not credited to battery ownership. The KPI does not subtract realized feedback wear from a different ideal schedule. Each counterfactual reports its own objective components. Directly subtracting a PV-only hindsight objective from the realized PV+BESS feedback profit would mix information and execution policies and is not the asset-value KPI.

This comparison is one-day operating value before capital expenditure, fixed operating costs, financing and taxes. A free terminal target can also use initial stored energy or leave different inventory; the selected 50% initial/terminal hindsight comparison avoids that particular inventory difference for the BESS model. Realized feedback may still finish at a different SOC. No residual-energy valuation, annualization, payback, NPV or investment optimization is implemented. Those need a representative horizon and additional cost/cash-flow assumptions.

Setting the switch to `false` skips the PV-only solve and its model, JSON, CSV and figure. Incremental-value metrics become `null` and an explicit baseline-enabled KPI is false. The perfect-information PV+BESS-versus-feedback comparison still runs. Four active auctions therefore produce four replay solves plus one PV+BESS hindsight solve and, when enabled, one PV-only solve. Counterfactual solve diagnostics are separate from replay solve counts.

13 Configurable participation and matched-day profile comparison

Participation is a choice of a nonempty subset P of the chronologically ordered auction set $\mathcal{A} = \{\mathrm{DA}, \mathrm{IDA1}, \mathrm{IDA2}, \mathrm{IDA3}\}$. The single-study configuration is `gme_oracle_experiment.participating_auctions`. Omission or null selects every configured auction dataset. Gate order remains unchanged; unselected auctions have no price-data requirement, trade variables, submission event/solve or fees. Their orders are absent rather than independent variables fixed after a solve. The active set on day d also removes documented/configured cancellations. If every selected auction is documented/configured as cancelled, the profile retains its full delivery day with zero orders, auction/control solves and auction fees. The zero-position feedback policy stores PV within the SOC box and curtails remaining surplus; executed wear and settlement remain in actual profit. There is no initial auction objective (its expectation KPIs are null), and no auction plan enforces the configured terminal target on that day. Missing or partial price data without cancellation evidence still fail.

Four independently optional auctions produce

$$|\{P \subseteq \mathcal{A} : P \neq \varnothing\}| = 2^4 - 1 = 15. \tag{67}$$

If DA is mandatory and the three IDA auctions remain optional, there are instead

$$|\{\{\mathrm{DA}\} \cup S : S \subseteq \{\mathrm{IDA1}, \mathrm{IDA2}, \mathrm{IDA3}\}\}| = 2^3 = 8. \tag{68}$$

Both families and explicitly listed profiles are supported. For example, $P = \{\mathrm{IDA1}, \mathrm{IDA2}\}$ and $P = \{\mathrm{IDA3}\}$ are valid without DA. All physical, native-product, degradation and settlement constraints continue to apply to the same full delivery day.

If the first selected gate occurs after delivery begins, its prefix uses zero contracted grid delivery and grid feedback with zero planned battery commands. The controller may store PV within the energy box and curtail remaining surplus; it does not earn uncontracted export revenue in this prefix. Actual energy and wear carry to the first selected solve, including the existing unfinished-interval state projection. No extra auction optimization is inserted before the first selected decision. Prefix executed/projected costs are subtracted from the first residual objective when forming an initial whole-day expected profit; the residual model objective is separately retained. Actual whole-day profit already includes all executed prefix costs through the ordinary financial accounting.

13.1 Sequential current-auction-only baseline

At the decision for auction a, integrated planning includes the open auctions $\mathcal{A}_a$. The configurable sequential baseline replaces this set by $\{a\}$. Let $Q_t^{<a}$ denote all previously submitted, fixed quantities, including unpriced commitments. Its only free market quantity is $q_{a,t}$:

$$Q_t = Q_t^{<a} + q_{a,t}, \qquad \max \sum_t \lambda_{a,t} q_{a,t} - C_{\text{deg}} - C_{\text{import}} - C_{\text{fees}}. \tag{69}$$

The existing physical balances link Q_t to grid delivery. The full remaining delivery horizon retains charge/discharge, curtailment, SOC transitions, native product equalities, POI limits and the configured terminal target. No later auction trade or revenue appears. Sunk earlier cash flow does not affect the choice. Each next gate receives updated PV, actual carried battery state and immutable prior submissions. Under the price-oracle convention, the current auction's actual clearing curve is supplied to that solve.

`gme_oracle_experiment.auction_planning_mode` selects `integrated` (default) or `current_auction_only`. `participation_study.sequential_all_auctions_baseline` optionally adds one all-auction sequential result beside the integrated participation profiles. The completed January 2025 noon-boundary refresh comprises 465 integrated profile/day cases and 31 sequential cases; its exact run identifier and results are retained in the evidence appendix. Matched daily realized profits, their paired differences, model sizes, forecast inputs and fallback decisions are saved. A later infeasible sequential solve retains prior physical commands and submits zero new trades; it has no previously planned future order to submit.

13.2 Operational fallback after an infeasible auction decision

Exact planned delivery does not guarantee feasible recourse after forecast updates. Limited remaining energy, power and trading capacity may conflict with the configured terminal state or remaining delivery obligations. The physical controller can continue within SOC/power limits by accepting an imbalance even when the planning model is infeasible. A pre-noon delivery shortfall is executed and settled under the retained policy; those closed intervals are excluded from the IDA3 MILP and cannot make it infeasible.

The configurable policy `execution.infeasible_auction_policy` defaults to `error`. The full-January study uses `previous_plan`. When and only when the solver confirms infeasibility, this policy uses the last feasible deterministic snapshot for the current auction's previously unsubmitted orders and remaining requested physical commands:

$$q_{a,t}^{\text{submit}} = q_{a,t}^{\text{last feasible}}, \qquad u_t^{\text{requested}} = u_t^{\text{last feasible}}. \tag{70}$$

The orders are submitted at their actual gate and priced at actual clearing prices with ordinary fee accounting; existing commitments remain fixed. The same feedback controller executes against actual PV, respects the physical limits, and settles unavoidable deviations ex post. This is a retained plan, not a new feasible optimum under the updated forecast. No recovery optimization or constraint relaxation is introduced. Subsequent selected decisions still attempt their normal reoptimization. Missing prior plans, nondeterministic snapshots and other solver failures remain errors.

Failed decision/forecast diagnostics, reused plan sources, submitted quantities and attempt counts are saved separately from successful solves; optional IIS reports use `execution.save_infeasibility_diagnostics`. Summary tables retain fallback-day and failed-decision counts rather than omitting those dates from the comparison. The physical trajectory can finish away from the terminal SOC target of its retained plan, just as under ordinary forecast-error feedback.

The `experiment-participation` command runs configured profiles on a common set of delivery days $\mathcal{D}$. The ranking metric is

$$\overline{J}(P) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} J^{\text{real}}(P, d), \qquad P^{\star}_{\mathcal{D}} \in \arg\max_{P} \overline{J}(P). \tag{71}$$

It is an empirical ranking on $\mathcal{D}$, not an established optimum for unseen days. Selecting a different winning profile after each observed day would evaluate $|\mathcal{D}|^{-1} \sum_d \max_P J^{\text{real}}(P, d)$ and would use additional hindsight; the saved daily-winner table is diagnostic and is not the fixed-profile ranking.

Every profile must complete exactly the same requested dates with fully priced actual settlement. Missing or invalid inputs fail the batch; they do not silently change a profile's averaging sample. Battery and operator fee initial state are identical across profiles and reset independently each study day. This is not a continuous multi-day carry of SOC or monthly traded volume. The initialization weather cutoff is common before the earliest candidate gate and delivery start. Later forecast updates occur only at selected auction decisions. Delivery grids, initial PV forecasts, effective realized PV (including configured fallback intervals) and shared actual auction-price curves are fingerprinted and checked for equality across profiles. The market adapter resolves native auction-product durations for each date. Actual price snapshots, model inputs and all daily decisions remain in the per-profile run artifacts.

Summary outputs retain average, total, minimum, maximum and median daily profit, 10th/90th percentiles, daily population standard deviation, negative-profit and zero-active-auction day counts, daily wins including configured tolerance ties, paired differences to the largest selected profile, mean SOC/energy, curtailment, wear, transaction fees and active auction/solve counts. Final stored energy is reported without salvage valuation. The current configured terminal SOC remains a planning constraint rather than a guarantee on feedback's final state, so cash profit and ending inventory must be assessed together. Fee and inventory reset assumptions also matter when extrapolating to a longer investment horizon.

The comparison figure places a separate gold perfect-foresight reference above the ranked profiles. Its mean and daily standard deviation use the full-day perfect-information objective values (Gurobi `ObjVal`) on the identical dates, effective realized PV, assets, costs and configured terminal SOC, with every configured auction that remains available. This reference includes markets outside a restricted candidate/profile family. Existing all-auction benchmark outputs are reused; if that profile is absent, one additional perfect-information solve per day supplies the reference without another auction replay. Its daily objectives and summary are retained separately, along with objective-minus-profit differences in the profile tables. This is a model-objective reference rather than an additional feedback participation rank: actual imbalance settlements and realized terminal inventory can differ between the accounting cases. `participation_study.perfect_foresight_reference` enables this feature by default and can be set to false.

The batch YAML section `participation_study` controls candidate/required auctions, explicit profile lists, common dates, output paths, per-profile figures/models, summary plotting and profit-tie tolerance. The supplied example is `configs/full_experiments/gme_participation_2025-01-01_03.yaml`. The full-month independent-day configuration is `configs/full_experiments/gme_participation_2025-01.yaml`, with 31 common delivery days and 465 profile/day cases. Fully cancelled IDA3-only cases remain in the averaging sample rather than changing that profile's denominator. The driver is `src/simulation/participation_study.py`, while `src/participation.py` validates/enumerates subsets. Summary CSV/JSON/plots are stored apart from each individual replay under its profile/day directory. Optional `participation_study.resume` verifies the same effective configuration, completed child manifests, input fingerprints and saved profits before reusing checkpoints. Incomplete children are retained separately and rerun; changed configurations are rejected. After independent solution validation, tolerated negative floating-point command roundoff is floored to zero before the physical controller's strict nonnegative-input checks.

14 Continuous operation with daily optimization horizons

14.1 Delivery books and the chronological event sequence

The separate `experiment-continuous` workflow selects inclusive local delivery dates $d \in \mathcal{D} = \{d_0, \dots, d_F\}$ through `continuous_optimization.start_date` and `end_date`. Each day has its own physical grid $\mathcal{T}_d$, native auction products, actual price curves and auction calendar. Commitments and revisable plans are keyed by delivery day as well as auction and interval, for example (d, a, t); the same auction name on another delivery day is a different market book. This is continuous chronological execution with daily planning horizons; it is not a single multi-day MILP, and a later-day solve cannot revise an earlier delivery book.

All events are merged into one chronological sequence. On an ordinary operating day d, the replay submits IDA3 orders for d, then DA, IDA1 and IDA2 orders for $d+1$ while executing delivery d. Submissions use the configured decision buffer before the local gate; price publications book already submitted orders without triggering another optimization. IDA3 variables remain restricted to noon–midnight of their delivery day. Native DA-hour equalities and quarter-hour IDA/physical controls are unchanged. Cancelled auctions produce no order, variables or solve, and unexplained missing inputs fail preflight.

This is a chronological sequence of *daily* MILPs, not a simultaneous month-long or two-day trading MILP. An optimization for $d+1$ cannot change current-day dispatch or commitments for d. Conversely, current-day IDA3 cannot reopen submitted next-day DA quantities. The current-day commands and book enter next-day planning through the projected boundary state below.

14.2 Forecast-based projection of the next initial state

For the default retained-plan controller, at a decision cutoff c_k, let x_k^{obs} be the most recent observed physical state at boundary $\tau_{i_k} \le c_k$. This state includes energy, cumulative executed throughput/degradation and, when relevant, depth-model segment history. The current continuous controller requires zero configured telemetry latency, treating an interval's actual state as observable at its end; preflight rejects a nonzero `source.telemetry_latency_minutes`. This restriction is independent of the weather forecast's availability lag. An unfinished interval's already issued command is retained. Let $\bar{u}_i$ be the retained current-day charge, discharge and PV-curtailment commands and Q_i^{sub} the cumulative submitted grid position. For a pre-delivery DA, IDA1 or IDA2 solve for d, the planning boundary $b(d,a)$ is midnight at the start of d; for IDA3 it is noon on d. Define a shadow state using only the weather/PV vintage available at c_k:

$$\widehat{x}_{i_k|k} = x_k^{\mathrm{obs}}, \tag{72}$$

$$\widehat{x}_{i+1|k} = \Phi\Big(\widehat{x}_{i|k}, \bar{u}_i, Q_i^{\mathrm{sub}}/\Delta_i, \widehat{P}_{i|c_k}\Big), \qquad i_k \le i < b(d,a). \tag{73}$$

Φ is the same grid-tracking and physical-protection controller used for actual execution. Thus the forecast projection respects SOC, battery power and POI limits while preserving the old commands and submitted grid target. It uses no new optimization for the locked prefix and no future measured PV. Forecast eligibility uses the UTC publication/availability clock at c_k, including the configured six-hour lag, independently of the later projected delivery boundary.

The daily MILP fixes its initial state to $x_{d,b} = \widehat{x}_{b(d,a)|k}$. At a later gate this projection is recomputed from the newer actual state and latest eligible forecast. Meanwhile actual execution advances separately:

$$x_{i+1}^{\mathrm{real}} = \Phi\Big(x_i^{\mathrm{real}}, \bar{u}_i, Q_i^{\mathrm{sub}}/\Delta_i, P_i^{\mathrm{real}}\Big). \tag{74}$$

Neither solving nor revising a shadow projection overwrites actual energy or aging state. The realized midnight state may differ from the previously projected state; later solves carry the actual

state as it becomes available. MPC instead uses the deterministic open-loop procedure in Section 8, supplying one state to the same auction MILP. Saved boundary records retain prediction, actual state and signed error $E_b^{\text{real}} - \widehat{E}_{b|k}$. Midnight projections are distinguished from IDA3 noon projections in diagnostics and plots.

14.3 Daily free decisions, initial condition and final target

For integrated planning, the free trade quantities are those of the current and remaining active auctions *for that delivery day*. For sequential `current_auction_only` planning, only the current auction has free trade variables and expected trading revenue. Previously submitted quantities for the target book, including unpublished pending orders, remain fixed in $Q_{d,t}^{\text{fix}}$ in both modes. Their cash values remain external to the new decision objective. Physical controls and SOC over the target day's remaining eligible delivery horizon remain free under the existing balances, costs and asset constraints. Trades in other delivery books are not decision variables in this MILP. Its objective is the daily residual objective (22); projected costs already locked before its planning boundary are not new trading opportunities.

For the supplied January configuration, $d_0 = 1$ January 2025 and $d_F = 31$ January 2025. The supplied boundary is

$$E_{d_0,0}^{\text{real}} = 0.90\, E^{\text{nominal}}. \tag{75}$$

Trading starts with the 31 December DA/IDA1/IDA2 events for delivery on 1 January. Their initial midnight state is this supplied assumption; it is not inferred from unavailable 31 December physical operation. Eligible 31 December forecasts and the actual 1 January delivery-price curves are still required. After the first midnight, there is no daily SOC reset.

The configured terminal constraint is applied only when planning the final delivery day and only when the target is non-null:

$$e_{d_F,|\mathcal{T}_{d_F}|} = E^{\text{nominal}}\, s_F/100, \qquad s_F = 50 \quad \text{(SOC percentage in the January example)}. \tag{76}$$

Under the default retained-plan controller, intermediate midnight states have no equality target or energy salvage price. Optional MPC instead uses its configured inventory treatment from Section 8. They remain inside their SOC box and, when configured, satisfy the additional planned minimum in (16). The January 2025 template leaves that minimum disabled; the paired January 2026 studies use 40%. Actual terminal energy remains an output of feedback execution and need not equal the planned target. The last delivery interval ends at local midnight on 1 February. The final day's IDA3 and all other auctions delivering on 31 January are included. On 31 January the DA, IDA1 and IDA2 auctions for 1 February delivery are outside the study and are not submitted or counted in January profit.

14.4 Independent trajectories, accounting and restart

The default experiment executes three independent chronologies: integrated planning, sequential current-auction-only planning and an integrated perfect-PV reference with the same daily horizons. All start from the same supplied state and use matched actual price, settlement, effective measured PV and asset inputs, but each carries its own decisions and realized state. The reference replaces forecast PV by effective realized PV, including in shadow projections. It does not introduce future delivery-day trade variables. Its reported performance is continuous settled operating profit, not the sum of overlapping auction objective values. The two trading modes and this reference are independently configurable.

The reference defaults to one optimization per delivery day, at DA (or the first active auction if DA is cancelled). The flag `continuous_optimization.perfect_pv_reoptimize_at_each_auction`

defaults to `false`; setting it to `true` restores optimization at every active gate. With the default, all later auction quantities are read from the initial integrated plan and submitted only at their own gates, while its physical commands remain in force throughout delivery. Publication, fees, actual state carryover and settlement follow the same chronological rules. The event trace identifies each retained submission separately from solver attempts. The initial plan is a perfect-PV planning solution at its modeled boundary, but the execution controller can alter it as actual SOC and retained curtailment recovery evolve. Its realized profit is therefore an information benchmark, not a guaranteed upper bound. Repeated solves can select different schedules when several optima exist, and finite MIP tolerance can permit small objective differences.

Monthly operating profit is additive over delivery products:

$$J_{\mathcal{D}}^{\text{real}} = \sum_{d \in \mathcal{D}} \left[\sum_{a,t} \pi_{d,a,t}^{\text{real}} q_{d,a,t}^{\text{sub}} + \sum_{t \in \mathcal{T}_d} S_t - D_d^{\text{executed}} - C_d^{\text{import}} - C_d^{\text{trading,real}} \right]. \tag{77}$$

The trade ledger preserves submission/publication clocks; daily profit is attributed to delivery date. Fees and auction revenues reconcile both by day and over the complete ledger. Operator fee volume still uses the submission month: the first day's 31 December orders belong to December, even though their revenue appears in 1 January delivery profit. Actual fee mode therefore requires an applicable December tariff and a timestamped initial operator volume state; January volume starts according to the fee calendar, without a daily reset. Wear is charged only on actual executed throughput/history. Per-day initial expectations and realized-minus-expected differences are reported separately; no objective values across overlapping gate solves are added to obtain profit.

With the configured infeasibility policy, a later failed solve can retain the previous physical plan and its already planned current order in integrated mode; the sequential mode retains dispatch and submits zero new trades. No prior plan means the existing failure behavior applies. Actual feedback still enforces physical limits and settles remaining deviations. Failed attempts and fallback choices remain explicit diagnostics.

Resume checks the effective configuration, explicit policy version and common input fingerprints before restoring an atomic completed-midnight checkpoint. The checkpoint contains actual physical/aging state, outstanding priced and unpriced commitments for both relevant books, retained commands, fee state, publication status and the event cursor. Work after that boundary is replayed without double-booking cash or fees. Business decisions rolled back for resume do not survive as submitted history; every actual Gurobi attempt, including discarded work, survives in a separate attempt log with its rollback marker. Native `Model.Runtime` statistics include these attempts, keep reference solves identifiable, and report totals plus mean/minimum/maximum/population standard deviation by strategy and delivery day. Model-build and Python wall times remain separate. Resolved configurations, common-input provenance, daily/monthly KPIs, state projections and comparisons are saved by file type under `outputs/continuous_optimization/farm_<id>/<run_id>/`. The manifest also records the Git revision, but the resume identity is not a hash of every source file. An incompatible policy-version change rejects an older checkpoint; the current exact-PV envelope version is distinct from the preceding controller version.

15 Implemented trading experiments and evidence

The independent-day historical runner is `experiment-gme-oracle`. It reads local cataloged market data, uses the same realized auction price curves for all strategies, forecasts PV using the configured cutoff-available weather, and executes the schedule against measured AC production. Supplying the future realized auction curves at the first decision is an intentional price-oracle assumption: this isolates physical/participation decisions but is not a causal price-forecast backtest and cannot be interpreted as implementable expected profit.

The implemented comparisons are:

- Integrated planning across the currently open auctions versus `current_auction_only`, where only the auction closing now has trade variables. Both carry the same submitted commitments as fixed positions.
- Independent-day participation profiles, comparing all nonempty subsets of DA, IDA1, IDA2 and IDA3 on matched dates, plus a sequential all-auction baseline. Cancellations create neither a solve nor trade variables.
- A frozen-plan replay that measures the value of later forecast updates and replanning while holding previously submitted trades fixed.
- A perfect-PV full-information schedule, and an optional PV-only schedule that removes the battery while preserving the market and farm assumptions. These are operating counterfactuals on the same realized prices and PV data; terminal inventory is reported separately because no terminal energy salvage value is included.
- Continuous daily planning across consecutive delivery days. Each strategy carries actual SOC, degradation history, submitted commitments, transaction-fee volume and cash ledger through local midnight. The daily MILP remains one delivery day at a time; it is not a month-long joint optimization. The January 2026 and August 2026 four-farm configurations compare updating deterministic and 51-member exact-delivery PV forecasts with a daily DA-only perfect-PV reference. August is restricted to 2–31 August because the required 31 July ensemble cycle for 1 August is absent. The opening SOC on 2 August is a new configured initial assumption, not a state carried from 1 August.

The January 2025 participation study uses independent delivery days. The continuous configurations expose inclusive start/end dates and initial SOC; January 2025 starts with DA submission on 31 December, supplied initial SOC 90%, and configured final planned SOC 50%. The January 2026 comparison configs also allow a 40% planned terminal floor. Those are study assumptions, not measured initial conditions or a guarantee on realized terminal energy. The perfect-PV continuous reference solves once at DA (or the first available auction) by default, then retains the plan; a shared flag can request reoptimization at later gates. It is an information benchmark, not a proven upper bound on realized feedback profit: actual controller actions, fees, settlement and final inventory can differ from the planned objective.

Every solve and interval records forecast vintage, position ledger, planned and executed battery flows, SOC, imbalance, throughput/wear, fees and timing. Outputs are grouped by run and file type under `outputs/`. Daily profits are reconciled from submitted trades, actual settlement, executed degradation and fees. The objective values of overlapping auction solves are never summed to represent profit.

16 Price forecasting integration status

The main-tree GME and continuous historical comparisons still use actual auction curves as oracle inputs. Its prepared-input route accepts explicitly timestamped point-price forecasts; its generic scenario tree accepts uncertain prices. These interfaces alone are not historical evidence of forecast-based trading value. The feature tree has completed offline DA/IDA1 forecasting benchmarks, and the MPC research tree has fitted chronological price models and completed explicitly forecast-based auction cases. Its physical economic controllers use forecast settlement-price paths. This is an implemented extension, not merely future work. The newer native-resolution candidate searches remain separate from portable MPC artifacts: a low validation error does not automatically install a model in the trading replay. No learned joint distribution of all auction spreads, PV and settlement

prices is established. Section 21 gives the formulations, results and information limitations of both price workflows.

17 Optional optimized-imbalance formulation

This section describes supported code that is disabled by the shared default. When `optimize_imbalances=true`, replace (21) by

$$\delta^+_{s,t} - \delta^-_{s,t} = g_{s,t}\Delta_t - Q_{s,t}, \tag{78}$$

$$0 \le \delta^+_{s,t} \le M_t y_{s,t}, \tag{79}$$

$$0 \le \delta^-_{s,t} \le M_t(1 - y_{s,t}), \qquad y_{s,t} \in \{0,1\}. \tag{80}$$

Here $M_t = (P^{\text{imp}}+P^{\text{exp}})\Delta_t$, optionally reduced by a configured imbalance-energy cap. The common M_t is valid because physical and contractual net energy use the same directional POI range. The sign binary prevents simultaneous positive/negative deviations, including with negative prices. Add to each scenario's objective

$$\sum_t \left[\lambda^+_{s,t}\delta^+_{s,t} - \lambda^-_{s,t}\delta^-_{s,t} - \kappa(\delta^+_{s,t} + \delta^-_{s,t})\right]. \tag{81}$$

If this mode has zero settlement price, zero penalty and no useful delivery restriction, an optimized contract need not be backed by delivered energy. The January farm-1 scenario study includes configurations with zero assumed planning settlement prices and illustrative absolute-imbalance penalties of 100 and 500 EUR/MWh. Their expected objective averages these terms over members; actual execution uses measured PV and settles deviations at recorded prices. The illustrative planning penalty is not a real market charge. The optimized-imbalance formulation retains all scenario leaves rather than using the exact-delivery envelope reduction; it is a mathematical study mode, not an assertion that deliberate imbalance strategies are permitted.

18 Data provenance, outputs and reproducibility

Farm inputs are local read-only Zarr/Parquet stores. API and workbook records use `data/api/market.duckdb`. Raw/normalized snapshots, source identities, checksums, revision and retrieval metadata support repeated offline studies. Historical publication remains a separate field/convention. Provider acquisition is an explicit user-run command; neither the MILP builder nor the historical experiment downloads missing inputs. DuckDB writers and analysis processes must be coordinated; complete cached days can be studied before an entire acquisition window is complete.

YAML controls study date, calendar validity, source mappings, PV and data policies, BESS and POI characteristics, state, degradation, solver and outputs. One include is resolved relative to the containing YAML, with recursive mapping overrides. Every shipped entry point imports `configs/standard.yaml` directly; experiment YAMLs define workflow-specific settings and overrides. The standard contains shared operational settings. Explicit null values and replacement lists survive the merge. The `resolve-config` command prints the effective YAML without running a study or accessing provider data.

A single root `farm_id`, optionally overridden with `-farm-id`, selects forecast and physical plant identifiers through typed references resolved after merging. Each farm uses its own weather and measured-power series and Parquet metadata for PV geometry and tracking. Automatic directional POI ratings use declared AC nameplate capacity divided by 1000 as an illustrative MW proxy; per-farm ratings or explicit plant limits override that rule. Battery size and SOC percentages remain common, while automatic battery dimensions resolve from `farms.<id>.battery` for the selected

physical plant. Numeric experiment overrides take precedence over these automatic dimensions. The reusable `configs/full_experiments/gme_day.yaml` exposes this workflow.

The current 50% study is configured by `configs/full_experiments/gme_oracle_2025-01-01_farm910_perfect_pv_comparison_terminal_soc_50.yaml`. Its inherited assumptions include a 5 MWh battery, 2.5 MW symmetric power, a 7–93% SOC box, directional combined efficiencies of $\sqrt{0.90} = 0.9486832981$ and a 6.638 MW directional POI. Their product gives 90% AC-to-AC round-trip efficiency for all farms. These are reproducible study settings, not confirmed anonymous-farm hardware.

The shared per-farm hypothetical BESS dimensions are:

Farm	PV AC nameplate (MW)	Battery (MWh)	Symmetric power (MW)
1	0.95328	0.75	0.5
345	1.5246	1.0	0.5
901	4.1600	2.0	1.0
910	6.6380	5.0	2.5

The shared 7–93% SOC limits are converted to energy after selecting capacity; replacement cost scales with that capacity. Throughput economics remain the configured historical USD 115/kWh proxy, 1.0389 USD/EUR conversion and 2,500 reference cycles at 80% depth. They have not been updated to a 2026 equipment quote. With the current efficiencies the derived AC-throughput wear cost is 27.635145 EUR/MWh; no dynamic capacity fade is simulated.

Fresh experiment runs default to `experiment_outputs.overwrite_existing:true`: repeating the same ID replaces artifacts within that run's scope under its configured output root. `resume: true` takes precedence and preserves checkpoints, requiring matching configuration and input fingerprints. Continuous output replacement holds the run lock and retains its inode; unrelated run IDs are preserved.

Each historical run is grouped under `outputs/full_experiments/gme_oracle/<run>/farm_<id>/<day>/` with `plots/`, `csv/`, `json/`, and optional `models/`. The effective configuration, input provenance and artifact manifest accompany auction plans, immutable ledger, actual dispatch, used PV vintages, settlement and objective comparisons. The main KPI distinctions are:

- `total_gain_eur`: actual feedback profit after available settlement;
- `perfect_information_objective_eur`: achieved full-information PV+BESS objective;
- `pv_only_perfect_information_objective_eur`: achieved optional PV-only objective;
- `battery_incremental_operating_value_eur`: the difference in (66);
- initial expectation, realized-minus-expected revenue/profit, and separate executed versus counterfactual degradation components.

Figures preserve delivery boundaries, including next-day local midnight. PV power curves visually connect successive interval samples; other interval powers and prices are drawn as held values, and SOC uses the actual boundary states. These presentation choices do not change the interval data or energy accounting. Snapshot settlement diagnostics include their different delivery windows and common-horizon comparisons.

Solver settings use relative MIP gap 10^{-4} (0.01%) by default, with no absolute MIP gap configured. Configurable `feasibility_tolerance` and `integrality_tolerance` map to Gurobi's numerical feasibility and integer tolerances. The historical YAML sets both to 10^{-8} to avoid tiny binary-mode leakage while preserving the independent validator's physical checks. The shared time limit is 120 seconds per solve with 10 solver threads. LP model files are not generated by default;

a study may explicitly enable model export, and infeasibility diagnostics have their own switch. Objective comparisons use achieved `ObjVal`, not a claim of exact global optimality beyond the recorded solver tolerance. Status, relative gap, model sizes and timing remain recorded. Before a schedule is used, independent validation recomputes physical balances, limits, native product consistency, information sharing, degradation and objective components. For optimized imbalance, the modeled cash flow, penalty and CVaR are independently recomputed from the solver's positive and negative imbalance variables. Those variables are also checked against the physical grid-minus-position balance at every interval. The validation report records the maximum EUR difference between split-variable and physically reconstructed accounting, without relaxing the energy-balance tolerance when a large penalty scales small MWh residuals. For continuous replay, the reported solver objective is checked against the independently recomputed objective using the absolute validation tolerance plus 10^{-8} times the larger objective magnitude; this relative term is distinct from the MIP gap and applies to that scalar comparison. It can be set via `execution.validation_objective_relative_tolerance`. Continuous objective components and, when available, the objective expression evaluated at the extracted solution use an absolute EUR allowance plus a separate relative allowance, checked for each scenario. The report records the applied tolerances and objective residuals. Ledger and physical reporting use the same configured absolute and relative numerical comparison convention. Nonfinite values, missing inputs and temporal/identity contracts still require exact validity. The solver's reported objective is not replaced by the recomputed value. In continuous replay, if the extracted objective expression and all independent physical and component checks pass, a discrepancy in the solver's reported objective is saved and displayed as a warning rather than stopping the replay. A missing or mismatched extracted expression does not receive that exception. When CVaR is enabled, the returned threshold and excess variables are checked against the scenario-loss epigraph and used to reconstruct the objective, including for a time-limited incumbent. Execution separately checks actual SOC and power/POI protection. Software consistency does not resolve uncertain farm metadata, forecast bias or historical publication vintages.

19 Current limitations and extension boundaries

The main-tree historical workflows use daily perfect-price planning with either one deterministic weather trajectory or an optional ensemble, in independent days or consecutive continuous operation, with approximate geometry and an idealized interval-average controller. Ensemble weather members are coherent input paths, but they are not calibrated probabilistic guarantees; exact scenario feasibility does not guarantee feasible delivery under unseen PV. Measured AC is effective observed production, not an independently estimated unconstrained irradiance-to-generation potential; hidden plant curtailment/faults in source production are not identified automatically. Configured cleaning and fallback choices remain part of the experiment.

The model currently omits continuous trading, order acceptance, market impact, multiple-farm network coupling, ancillary services, intrainterval dynamics, ramping/startup constraints, dynamic capacity loss, calendar aging, thermal battery dynamics, fixed operating costs and investment cash-flow optimization. The generic `backtest` CLI remains synthetic-only; the separate historical `experiment-gme-oracle`, `experiment-participation` and `experiment-continuous` workflows are the implemented manual-data replays. The sibling research tree additionally evaluates explicit forecast-price cases. Their historical clocks remain assumed; they do not retroactively turn main-tree oracle results into causal backtests. A joint multi-auction price/PV scenario model and a broader selection-fixed economic evaluation remain open.

The original implementation-status report is retired. This source and its PDF are the maintained mathematical description; run manifests and saved solver results are the numerical evidence for particular configurations. Practical commands and configuration examples are in the root `README`

.md.

20 PV forecasting: experiments, formulations and first results

20.1 Targets, information and the meaning of a forecast error

Four anonymous farms have AC capacities 0.95328, 1.5246, 4.16 and 6.638 MW. Location, some geometry and every hypothetical battery/POI parameter remain modeling assumptions. A weather initialization, its assumed release, the target interval, the time a target becomes observable, and the time a fitted model becomes available are different clocks. For a decision at k, admissibility is

$$A_v \le k, \qquad A_{\text{model}} \le k, \qquad A_{\text{label}} \le k \quad \text{for any feedback used in fitting/updating.} \tag{82}$$

Historical studies can explicitly relax model availability, but must retain the retrospective label. The six-hour weather-release convention is assumed. An initialized forecast is not automatically available at initialization.

The reviewed dataset distinguishes retained measured AC, approved synthetic replacement, policy-defined zero, and still-excluded/missing intervals. A historical-GHI physical reconstruction estimates a counterfactual production profile; it does not establish that an observed low-output episode was faulty. The newer optimization profile fills retained daylight exclusions with its stored historical-GHI reference and fails on unresolved daylight gaps. The training and scoring masks are specific to each study. Scores on old raw/curated targets and newer reviewed targets cannot be compared as a controlled ablation.

For quarter t, normalized target is $y_t = P_t/P_{\text{AC}}$. With eligible daylight quarters $\mathcal{I}_d$ on day d, the reviewed search uses day-balanced scores

$$\text{nRMSE} = 100\sqrt{\frac{1}{|\mathcal{D}|}\sum_d \frac{1}{|\mathcal{I}_d|}\sum_{t\in\mathcal{I}_d}(\hat{y}_t - y_t)^2}, \tag{83}$$

$$\text{nMAE} = \frac{100}{|\mathcal{D}|}\sum_d \frac{1}{|\mathcal{I}_d|}\sum_{t\in\mathcal{I}_d}|\hat{y}_t - y_t|. \tag{84}$$

Signed bias replaces the absolute residual by $\hat{y} - y$. Day-balanced and quarter-weighted summaries need not coincide. A day's measured and synthetic subsets may both be present, so subgroup day counts need not sum to the all-target day count. Auction views overlap and cannot be summed as independent production.

20.2 Physical calibration and irradiance diagnostics

The original physical-calibration workflow minimizes a bounded, normalized day/auction loss, optionally regularized toward reference assumptions:

$$\min_{\theta\in\mathcal{B}} \frac{1}{|\mathcal{G}|}\sum_{(d,a)\in\mathcal{G}} \frac{1}{|\mathcal{I}_{da}|}\sum_{t\in\mathcal{I}_{da}} \rho\left(\frac{f(X_{dat};\theta) - P_{dt}}{P_{\text{AC}}}\right) + \lambda\sum_j \left(\frac{\theta_j - \theta_{0j}}{u_j - l_j}\right)^2. \tag{85}$$

Candidate families fit an AC scale, Faiman coefficients and, for trackers, ground-coverage ratio and maximum rotation; bifacial candidates additionally state rear-irradiance assumptions. Monthly training precedes the issue, the last seven completed days select history/transposition/regularization, and the selected candidate is refitted on the eligible history. The original 2025 study used January warmup and 334 February–December evaluation days. The earlier feature-tree LaTeX records held-out daylight RMSE reductions from 0.189674 to 0.076846 MW (farm 1), 0.333161 to 0.144849 (345), 0.632407 to 0.571855 (901) and 1.119012 to 1.032802 (910). These are retained historical report values; the full original physical-calibration run directory is not present in the inspected main-tree output inventory, so they are secondary

evidence rather than a newly reconstructed result. The final all-year parameter fits are in-sample and were treated as prior effective metadata. Current shared temperature defaults differ from those historical fitted thermal coefficients.

A later retrospective physical study used historical satellite GHI and the provider-confirmed historical diffuse fraction, forecast temperature/wind from four cycles, and a January 2025–July 2026 fitting period. August 2026 selected the configuration. It evaluated 624 initial trials and 12 refinements; bounded L-BFGS-B optimized thermal coefficients, temperature coefficient and scale. The four weather views each carried weight 0.25; they were correlated views, not four observed weather realizations. Daytime RMSE normalized by AC rating was:

Farm	Selected proxy	Selected configuration	Original	Selected
1	Milan	Unchanged reference	3.416%	3.416%
345	Genoa	Fixed, isotropic, Huber	13.493%	8.675%
901	Trieste	Rear-enabled tracker, Huber	8.192%	7.290%
910	Aosta	Tracker, isotropic, MSE	14.027%	13.151%

These are historical-irradiance conversion scores, not operational forecast scores. Several fitted coefficients reached bounds, structural trials were unbalanced, and August was used for selection. The saved report `docs/results/pv_realized_irradiance_2025_2026/README.md` preserves those qualifications. In a matched August 20–31 daylight forecast comparison, January-only quantile calibration gave new/previous-physical CRPS of 9.970/9.970, 8.901/9.692, 7.306/7.878 and 7.938/7.850 percent AC respectively. The unchanged farm-1 physics had identical January-only scores. Its much better earlier calibrated score used additional calibration history; it does not identify a physical-model improvement.

The older neural-pilot study trained on 2025, validated on January–April 2026 and inspected May–August. Timing ablations added lead and cycle hour, with 80 fits across five farm/family candidates. The retained TCN choices were partly informed by the later period, so they are research pilots. Satellite-GHI input swaps diagnose sensitivity to irradiance error; they leave other weather and unobserved operation uncertain. In particular, farm 910's August low-output episodes were not automatically diagnosed as faults by this exercise.

20.3 Completed reviewed-GHI model search

The completed run `historical_ghi_all_farms_night3000_v1` contains 8,192 candidate fits: four farms, two feature sets, sixteen families and 64 sampled settings per group. It completed 132 selected-model refits. Families cover regularized linear/robust regression, spline/polynomial/random-feature ridge, random/extra trees, histogram/LightGBM/CatBoost boosting, MLP, TCN, bidirectional GRU/LSTM and Transformer. This is a sampled search, not a global architecture optimum. Training used 2025, validation January–April 2026 and test May–August, with the saved boundary embargo.

Training substitutes historical GHI for forecast GHI, while other weather comes from the preceding delivery day's 00 UTC cycle. Validation/test use forecast GHI. Hourly mean radiation is repeated over its constituent quarters; instantaneous weather is interpolated to quarter-hour midpoints. Preprocessing is fitted on training only. Exactly 3,000 eligible physical nighttime training quarters per farm are selected across the year. Daylight performance controls selection; 2026 has already been inspected and contains synthetic labels.

For a neural predictor, let X be the available weather sequence and let $f_\theta(X)_t$ be its normalized AC prediction. The core learning problem is

$$\min_\theta \sum_{d,t} \omega_{dt}\, ho\big(f_\theta(X)_t - y_{dt}\big), \qquad \sum_{t\in d} \omega_{dt} = \frac{1}{|\mathcal{D}_{\text{train}}|}, \quad \hat{P}_t = P_{\text{AC}}\,\text{clip}(f_\theta(X)_t, 0, 1). \tag{86}$$

The selected family/hyperparameters define MSE or Huber loss; the neural Huber transition is 0.05 of AC capacity. AdamW trains up to 150 epochs with patience 15 on validation daylight error. The training objective and early-stopping criterion are separate from the final test score; there is no post-hoc calibration in this point-model search. Monthly night windows force zero output.

An MLP applies successive affine maps and nonlinearities to weather features. A TCN combines finite dilated convolutions over the supplied forecast sequence. A bidirectional recurrent model concatenates forward and backward hidden states of that same issued forecast. Transformer attention has the usual form

$$H = \mathrm{softmax}(QK^{\mathsf{T}}/\sqrt{d_k})V, \qquad Q = XW_Q,\ K = XW_K,\ V = XW_V, \tag{87}$$

with the configured positional, masking and feed-forward transformations. Using future *forecast* covariates in a sequence is not future-target disclosure when the complete forecast was issued by the cutoff. In contrast, using future observed GHI in validation would be disclosure and is excluded. Sequence boundaries and shortened inference context remain material choices.

The following validation-selected winners use extended features. Scores are day-balanced daylight percentages of AC capacity. Test ranking was not used to replace the validation winner.

Farm	Model	Validation	Test	Test MAE	Test bias
1	bigru	4.708	6.069	4.006	-1.048
345	transformer	7.645	10.366	6.893	-1.169
901	bilstm	10.241	11.231	6.776	1.137
910	tcn	8.977	10.992	6.630	1.071

Zero and a training-only day-balanced constant are included in the saved baseline table. The 132 saved selected models were subsequently rescored with nighttime clipping; daylight metrics and all 8,192 validation rankings were unchanged. The retained inference bundles are the four winners, not every historical fitted weight file. Test errors separated by target origin are:

Farm	Test target	Quarters	Days	nRMSE (%)	nMAE (%)
1	all	8094	123	6.069	4.006
1	measured	8065	123	6.067	4.004
1	synthetic	29	3	3.977	2.993
345	all	8068	123	10.366	6.893
345	measured	8058	123	10.364	6.888
345	synthetic	10	3	10.179	8.482
910	all	8075	123	10.992	6.630
910	measured	5735	90	12.045	7.326
910	synthetic	2340	41	8.702	5.615
901	all	8108	123	11.231	6.776
901	measured	8037	123	11.251	6.764
901	synthetic	71	7	6.695	4.613

Farm 910 has 2,340 synthetic scored test quarters; its all-target 10.992% nRMSE cannot be interpreted as independent metered accuracy. Its measured-only 12.045% and synthetic-only 8.702% describe different populations, not a paired estimate of reconstruction bias. Seven-day block bootstrap comparisons are conditional on the chosen targets and previously inspected dates.

20.4 Four-cycle, 48-hour PV extension in progress

The new neural study uses cycles at 00, 06, 12 and 18 UTC and 192 quarter-hour targets per 48-hour issue window. A physical target can occur in multiple vintages. All copies retain the target's

chronological split, and the first six hours precede the assumed release. These early leads can be retrospective training examples but are excluded from availability-eligible validation selection. The proposed training objective extends (86) to issue/lead pairs (v, ℓ) with $t = t_v + \ell$; the effective sampling weights are defined by the saved data preparation, not by assuming overlapping views are independent observations. Uncertainty reporting must block by target day.

The checked dataset has, per farm, 195,264 training pairs (including 24,000 views of 3,000 selected nighttime quarters), 47,656 validation pairs and 56,826 test pairs. Core weather must be present for a supervised row; additional gaps use training medians, with no missingness indicator channels in this design. Boundary sequence-context missingness remains reported. The five neural families, two feature sets and 64 draws make 2,560 candidate fits, followed by 40 family/feature refits and up to eight extra top-three refits.

The CPU, single-GPU, parallel-GPU and multi-GPU attempts have distinct identities. The multi-GPU search was interrupted by GPU-memory contention; the recovery record at this snapshot has 220/2,560 completed candidates and mode `gpu0_recovery`. Other saved partial attempts must not be pooled into a completed search. The configured parallel recovery uses two fit workers on GPU 0 and retains matching earlier records. No final four-cycle ranking or neural-controller result is available in the captured evidence. Synthetic GPU throughput checks are engineering measurements, not model-accuracy experiments.

20.5 Marginal ensemble calibration and delayed online CRPS

The completed January/August marginal study uses 49 training and 12 validation delivery days. Forecasts from 50 weather members are converted to PV; model selection separately ranks the member mean by RMSE, median by MAE, and calibrated quantiles by CRPS. For quantile level τ, the pinball objective is

$$\min_{\beta} \sum_{i} \rho_\tau(y_i - x_i^{\mathsf{T}}\beta), \qquad \rho_\tau(r) = r\big(\tau - \mathbf{1}\{r < 0\}\big). \tag{88}$$

Quantiles are bounded/sorted as prescribed by that study, linearly interpolated between saved levels, with constant tails beyond p05 and p95. This is a marginal distribution; it does not identify a trajectory law.

For expert predictions x_m and probability weights w_m, exact weighted empirical CRPS is

$$\mathrm{CRPS}(w, x, y) = \sum_{m} w_m |x_m - y| - \frac{1}{2} \sum_{m,j} w_m w_j |x_m - x_j|. \tag{89}$$

Quantile-curve CRPS instead integrates twice the pinball loss over probability. Finite-grid quadrature is not an exact empirical-ensemble score.

The delayed ML-Poly implementation uses normalized powers and

$$g_m = |x_m - y| - \sum_j w_j |x_m - x_j|, \qquad r_m = \sum_j w_j g_j - g_m, \tag{90}$$

$$R_m \leftarrow R_m + r_m, \qquad S_m \leftarrow S_m + r_m^2, \tag{91}$$

$$\eta_m = (1 + S_m)^{-1}, \qquad w_m^{\mathrm{next}} = \frac{\eta_m \, [R_m]_+}{\sum_j \eta_j \, [R_j]_+}. \tag{92}$$

A zero denominator uses uniform weights. Learners are separate by farm, auction and three-hour local block. Expert identities are sorted ranks: 50 ensemble-only experts, or 70 including a deterministic point and 19 error quantiles. Warm starts replay eligible earlier out-of-fold feedback; cold starts begin uniformly. Each issue freezes weights. Delayed gradients use the weights originally

issued, and feedback is released only after target availability. No regret theorem is claimed for this modified delayed replay.

On August 20–31, the DA CRPS comparison is:

Farm	Matched quarters	Equal ensemble	Fixed quantiles	Warm combined
1	1152	2.493	2.048	2.120
345	1152	2.774	2.418	2.704
901	1152	4.792	4.307	4.630
910	1101	2.793	2.288	2.503

Units are percent AC; these full matched samples include night. Fixed quantiles beat warm online combination at DA on all four farms. At IDA3, warm combination improves fixed-quantile CRPS for farms 1 (2.318 versus 2.403) and 910 (2.204 versus 2.352), but not 345 or 901. This is mixed exploratory evidence, not a general online-learning benefit. Candidates were selected on these validation dates; August 20 early auctions include a retrospective source-model case. The complete 112-row comparison is retained in the original metrics file.

The matched same-day-cycle study is a separate hindsight sensitivity: the current day's 00 UTC cycle is unavailable at that day's DA/IDA1 cutoffs. Calibrated CRPS changes from preceding-day to same-day were 2.05 to 1.89, 2.42 to 2.55, 4.31 to 3.86 and 2.29 to 2.09 percent AC for farms 1, 345, 901 and 910. Selection was repeated on the same 12 days, so this is neither an isolated vintage ablation with frozen models nor evidence of a tradable DA improvement.

20.6 Censored-distribution and quantile-network extension

The Mayer-style extension is implemented but no completed historical run directory was found in the inspected outputs. Its current broader configuration requires ensemble and target coverage beyond the available August boundary. It therefore has no new numerical result in this report. The input is the normalized hourly PV ensemble plus a deterministic control *proxy*; only complete valid four-quarter hours enter. The proxy is not an identified meteorological control member.

For $Z \sim N(\mu, \sigma^2)$, doubly censored PV is

$$Y = \min(1, \max(0, Z)), \quad \Pr(Y = 0) = \Phi(-\mu/\sigma), \quad \Pr(Y = 1) = 1 - \Phi((1-\mu)/\sigma). \tag{93}$$

Basic EMOS links location to predictors and log scale to ensemble variance, fitting per lead by CRPS. Boosted censored regression pools leads and fits likelihood. Linear quantile regression uses sorted members and the proxy. The neural censored-distribution model predicts $\mu_\theta(x)$ and $\sigma_\theta(x) = \text{softplus}(h_\theta(x)) + \epsilon$ and minimizes mean censored-normal CRPS. Ten seeded models average location and scale; this is not a ten-component mixture distribution. The analytic PyTorch score retains native gradients and is checked against R `scoringRules`; R `crch` and scoring run in isolated subprocesses [5].

QRNN predicts 51 quantiles using a smooth asymmetric quantile loss. A Bernstein quantile network imposes monotonicity through ordered coefficients:

$$Q_\theta(\tau) = \sum_{j=0}^{K} a_j(x) \binom{K}{j} \tau^j (1-\tau)^{K-j}, \qquad a_j = a_0 + \sum_{h=1}^{j} \text{softplus}(b_h). \tag{94}$$

The noncrossing network uses a positive triangular cumulative construction. Its architecture is an explicit implementation choice, not an exact paper reproduction. Out-of-range quantiles are flagged/excluded from matched scores rather than silently clipped in this post-processing study. The quantile families have 100 seeded random trials per farm, inner five-day blocks for training/stopping/selection and separate outer evaluation. No copula or MILP scenario coupling is produced by these marginal models alone.

21 Price forecasting across the two sibling workflows

21.1 Feature data and availability contract

The feature checkout builds snapshot-pinned market/fundamental/weather tables. Its 2024 coverage report contains 8,784 native hourly NORD DA labels with no missing quarter-hour-equivalent slots; load and total-generation forecasts also cover the year. Solar and wind each miss 388 quarter-hour slots. The French national balance misses four such slots at the autumn DST transition. Gas IGI has 366 daily values. Regional weather contains 438 of 732 requested cycles. These counts are from the saved coverage artifact, not a new database query. Historical publication/revisions remain unverified for several sources; weather includes hindcast or unverified archive records. A national electricity balance is not a physical cross-border flow, and an intraday NTC is not a DA capacity.

At an origin k, a usable feature is the latest source value with stated $A_j \leq k$, and training labels must also precede the fitting cutoff under the declared publication policy. The features benchmark explicitly assumes delivery-day IGI and ENTSO-E day-ahead fundamentals available before DA/IDA1; its strict alternatives reject unknown clocks. The initial MPC context study uses 48-hour shifts for French/Swiss realized observations and PSV, and a nine-day Swiss-border shift with an assumed weekly release. A separate previous-day experiment is intentionally more permissive. These conventions are scenario assumptions, not recovered historical publication evidence.

The base feature vector contains local calendar harmonics and eligible price lags. Fundamentals add NORD load, total generation, solar and wind forecasts. Context ablations add gas, Milan/Turin weather, French nuclear/hydro and commercial exchange, Swiss balance, border schedules/NTC/spreads, or reservoir information. Missing observations remain gaps. Native hourly prices are held flat when aligned to quarter-hours; interpolation must not invent quarter-hour target variation. The newer searches separate hourly and quarter-hour labels before expansion, unlike the explicitly coarse 2024 intraday training extension.

21.2 Point, spread and distributional prediction problems

The feature-tree tasks are DA before DA, IDA1–DA spread before DA, that spread before IDA1, and direct IDA1 before IDA1. Only post-DA origins may use target-day published DA prices. With $s_t = p_{1,t} - p_{DA,t}$,

$$\hat{p}_{1,t} = p_{DA,t} + \hat{s}_t \quad \Longrightarrow \quad p_{1,t} - \hat{p}_{1,t} = s_t - \hat{s}_t. \tag{95}$$

Thus the zero-spread baseline is the published DA curve and is an essential post-DA comparator. The newer MPC searches also model IDA2, IDA3 and settlement, use earlier-cleared same-delivery auction curves, and predict spreads relative to the most recently cleared auction where configured. Adjacent delivery slots of an already published earlier-auction curve are allowable inputs; later products of the auction being predicted are not disclosed labels.

For training-standardized predictors $\tilde{x}_i$, regularized regression solves

$$\min_{\beta_0,\beta} \sum_i \omega_i \rho(y_i - \beta_0 - \tilde{x}_i^{\mathsf{T}}\beta) + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2. \tag{96}$$

Loss/normalization follow each estimator: ridge uses squared loss and L^2; Lasso uses L^1 coefficient regularization; elastic net mixes penalties; Huber estimates robust location. The estimator-specific loss scale matters when comparing penalty parameters. Spline and polynomial/random-feature variants change the design matrix. Neural point models, trees and boosted trees use their specified squared, absolute, Huber or quantile losses. MAE-based selection does not imply that each model trained on MAE.

The current Huber candidate uses training-only target scaling as well as feature scaling and inverse-transforms predictions to EUR/MWh. Earlier Huber artifacts precede that change. Interaction

regressions augment noncalendar signals z with $z\sin(2\pi h/24)$ and $z\cos(2\pi h/24)$, allowing time-of-day effects while retaining a pooled model. Daily-curve inputs concatenate lagged curve features; delivery-slot fits are a separate representation. Window experiments restrict eligible training to the last 30, 60 or 90 days versus the full available history; they change fitting data, not the definition of the target.

In the feature benchmark, Gaussian and Johnson-SU networks minimize negative log likelihood. With $Z \sim N(0,1)$, the latter has

$$Y = \xi + \lambda \sinh((Z-\gamma)/\delta), \qquad \lambda, \delta > 0, \tag{97}$$

with constrained neural parameter maps. Its location ξ is not generally its mean; actual mean, median, quantiles and PIT are returned separately. Quantile models minimize pinball loss; mean/median/robust-location outputs are not silently interchanged. NBEATSx and a bounded local-checkpoint TabPFN pilot are optional benchmark architectures, not a new probabilistic trajectory law.

21.3 Residual trajectories, combinations and forecast scores

The initial MPC ridge workflow sets $\alpha = 20$ with a 14-day chronological calibration tail. A predictor fitted before the tail generates its residuals; the deployed point predictor is then refitted through the eligible fold cutoff. For complete local-day residual block $\varepsilon_{d_j,t}$,

$$\hat{\pi}_t^{(j)} = \hat{\mu}_t + \varepsilon_{d_j,t}. \tag{98}$$

This preserves empirical within-day error shape. DST matching is explicit; separate days are sampled independently. It does not learn cross-market, cross-day or PV–price dependence, and the finite scenario mean need not equal the point predictor. The basic studies completed 30 chronological fits for January/August and 27 extended fits including October and coarse 2024 IDA data.

An equal combination averages selected point forecasts. A convex combination fits earlier out-of-sample predictions F_{im} under

$$\min_{w\geq 0,\ \mathbf{1}^\top w = 1} \frac{1}{N}\|y - Fw\|_2^2 + \lambda\|w\|_2^2, \qquad \hat{y} = \sum_m w_m \hat{y}_m, \tag{99}$$

with scaling/regularization defined by the particular study. Size and penalty are chosen on an earlier tuning slice; the weights are then refit there before later evaluation. Local-hour-block combinations select separate pools by time block. QRA replaces the squared objective by quantile losses. Conformal helpers use a prior residual quantile for intervals; no universal conditional-coverage claim follows under drift. Feature-tree QRA/learned combinations are executable helpers without completed historical evaluation there; the MPC-tree temporal combination studies have completed saved evaluations.

For price observations and forecasts on a common sample,

$$\mathrm{MAE} = N^{-1}\sum_i |y_i - \hat{y}_i|, \qquad \mathrm{WAPE} = 100\frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i|}, \tag{100}$$

$$\mathrm{rMAE} = \frac{\sum_i |y_i - \hat{y}_i|}{\sum_i |y_i - \hat{y}_i^{\mathrm{naive}}|}. \tag{101}$$

Zero denominators produce undefined relative scores. The feature-tree benchmark duration-weights within days, then weights days equally; the native-resolution search reports pooled products. These weightings must not be silently merged. CRPS evaluates the distribution independently of the point score and cash profit. Finite quantile-grid integration, analytic Gaussian CRPS and exact empirical CRPS are labelled separately.

21.4 Completed feature-tree DA/IDA1 results

The January–March 2026 validation run `20261023T1330058671139Z` completed 90 days, four tasks and 94,996 forecast-statistic records, with no recorded failures. These are not 94,996 independent price observations: several statistics/models share each target. The study used an 84-day rolling window, saved refit schedule and assumed availability. Its April–August final-test flag is false in this workflow; related periods were later inspected elsewhere and are not globally untouched.

Task	Model/features	MAE	RMSE
DA before DA	Previous week	16.57	22.97
DA before DA	Elastic net, price/calendar F0	11.41	16.50
DA before DA	Elastic net, fundamentals+gas F2	8.11	12.57
IDA1 spread after DA	Zero spread	4.90	6.61
IDA1 spread after DA	LightGBM F2	5.02	6.66
IDA1 spread before DA	LightGBM F2	4.98	6.65
Direct IDA1 after DA	Elastic net F2	5.93	8.07

All errors are EUR/MWh, daily duration-weighted on the hourly representation. DA elastic net F2 improves paired MAE by 8.45 EUR/MWh versus previous week; the saved seven-day block interval is $[-12.06, -5.92]$ for model minus baseline. This is a combined feature/model contrast, not a gas-only causal effect. For post-DA IDA1, the zero-spread benchmark is better than these fitted models.

The separate January 5–11 challenger week uses matched dates and daily refits. DA/post-DA-spread MAEs were ridge 7.27/3.97, LightGBM 6.70/3.62, CatBoost 5.40/3.69, MLP 5.30/3.83, Gaussian network 6.44/3.70, Johnson-SU 5.32/3.95, NBEATSx 10.24/5.70, TabPFN 5.99/3.99, and equal ridge/MLP 4.91/3.82. Zero-spread MAE was 3.95. Seven days cannot establish a seasonal architecture ranking. Separate weekly-refit/full-profile tests gave pooled-versus-slot ridge DA MAE 12.20/12.87 and spread MAE 4.66/7.70. A matched F0–F4 feature ablation gave DA MAEs 10.74, 8.51, 12.20, 11.29, 11.29: more context did not uniformly help. These different refit/design settings prevent cross-run architecture attribution. The October 26 compact-design four-day pilot is also complete, but is a workflow check rather than a replacement for full-quarter validation.

21.5 MPC-tree price studies and selection-fixed follow-up

The first chronological ridge study reports the following representative point/distribution comparison on common days:

Period/market	Model	Days	MAE	CRPS
January DA	Price-only ridge	31	10.260	9.183
January DA	Fundamentals ridge	31	7.545	6.915
August DA	Price-only ridge	31	17.123	14.542
August DA	Fundamentals ridge	31	19.011	12.192
January settlement	Fundamentals ridge	31	47.670	35.065
August settlement	Fundamentals ridge	31	73.339	45.169

Fundamentals improve January point accuracy but worsen August DA point MAE while improving its CRPS. Distribution and point metrics therefore support different conclusions. In the separate European-context January DA ablation on 30 common days, price-only/fundamentals/combined MAEs are 10.427/7.715/7.285; coverage and training samples differ from the 31-day study, so those numbers are not a single controlled sequence of gains.

Expanded searches examine native resolution, daily curves, slot models, earlier-cleared auctions, spread targets, French/Swiss context, Terna load, recent windows and robust regression. All 40

captured run manifests, their status and saved record counts are listed in Appendix D; the accompanying CSV retains every saved candidate score, not just winners. The central later-evaluation evidence is:

Market	Earlier-selected model	Later MAE	WAPE	rMAE	Products
IDA1	Spread/combined LightGBM L1	6.88	6.10%	0.498	8160
IDA2	Cleared-only LightGBM L1	7.12	6.26%	0.472	8064
IDA3	Cleared/combined Lasso 1	10.54	9.08%	0.534	4176

These are the *first-ranked* candidates from June–August 2025 selection, followed on shared complete days from September 3–November 30 with 13 weekly refits and a one-day gap. Products and label publication must precede each fit. The full follow-up contains 27 fits, including other earlier-selected candidates and ridge references. Choosing its best row after inspection would be a new selection. Historical availability remains assumed and the broader research history prevents a globally untouched-test claim.

For native quarter-hour DA, previous-day IDA1/IDA2 prices at 24/48-hour lags were tested across 35 candidates and matched feature sets. The February–April validation minima were 9.46% WAPE with IDA inputs versus 9.64% without, on 82 common days. The earlier top candidate, fundamentals interaction ridge 200, was then followed on 107 complete days in May 3–August 31, with 18 weekly refits and 10,272 products: MAE changes from 16.314 to 16.021 EUR/MWh and WAPE from 11.23 to 11.03%. All ten matched follow-up fits completed. The small paired benefit is distinct from selecting a different model after seeing later scores.

Temporal combinations rank candidates on February 1–March 15, tune on March 16–31 and evaluate April 1–30, 2026. Equal top-ten ensembles give IDA1/IDA2/IDA3 MAEs 5.989/7.836/13.525 EUR/MWh and WAPEs 4.94/6.44/12.10%. Convex stacking gives 6.040/7.838/13.563; local-hour combinations give 5.995/7.809/13.525. More complex combination did not uniformly improve this previously inspected later slice.

Matched French-balance and Terna-load ablations show heterogeneous, often tiny median MAE changes across 27 candidates per task/feature pair. Swiss-reservoir median changes range from -0.004 to -0.017 EUR/MWh in the saved comparisons. Those exploratory multiple comparisons do not establish an economically meaningful causal benefit. Pairing training rows as well as test products is necessary to separate added information from coverage changes.

The 96-fit recent-window DA search is complete: its lowest validation WAPE is 9.01% for 90-day fundamentals interaction elastic net, versus 9.46% for the previous best full-history/IDA configuration. These are selected minima on the same period. The paired later rolling-window comparison is still recorded as running at capture, so no later advantage is asserted. The bidirectional earlier-auction-neighbor search is also incomplete; its offsets are -60, -15, 15 and 60 minutes, with saved boundary handling.

The first DA Huber search completed 56 fits but logged iteration-limit warnings; its scores are provisional. A target-scaled 1,000-iteration probe still warned. The 144-fit IDA Huber search has no warning in its relevant old shared-log part, but lacks candidate-level convergence evidence. Its minimum Huber WAPEs were 4.89/5.48/9.18%, versus Lasso minima 4.74/5.49/9.18%. Current code records warnings per candidate and excludes warned fits from ranking. Completed output is not proof of numerical convergence, and current warning handling cannot be retroactively assumed for old results.

22 Real-time research extensions: mathematical specification

22.1 Forecast-economic control and pathwise monetary risk

The MPC sibling extends the common-control model with settlement forecasts, adaptive PV-history recourse, dependence transformations, scenario reduction, continuation values and controlled information/asset experiments. Its launcher uses continuous replay: each case carries its own actual SOC, commitments, degradation and fee ledger. Fixed-ledger replay remains an older diagnostic.

Let $n(s,t)$ own controls and let $a_{n,t} \in [0,1]$ be PV utilization, avoiding confusion with the earlier curtailment-power variable u. Define

$$g_{s,t} = a_{n(s,t),t} V_{s,t} + d_{n(s,t),t} - c_{n(s,t),t}, \tag{102}$$

$$b^+_{s,t} - b^-_{s,t} = \Delta_t g_{s,t} - Q^{\mathrm{sub}}_t, \qquad b^{\pm}_{s,t} \geq 0, \tag{103}$$

$$R_s = \sum_t (\hat{\lambda}^+_{s,t} b^+_{s,t} - \hat{\lambda}^-_{s,t} b^-_{s,t}), \qquad L_s = D_s + I_s - R_s. \tag{104}$$

The split variables are bounded by $b^+_{s,t} \leq \max(0, P^{\mathrm{exp}}\Delta_t - Q^{\mathrm{sub}}_t)$ and $b^-_{s,t} \leq \max(0, P^{\mathrm{imp}}\Delta_t + Q^{\mathrm{sub}}_t)$. If directional prices differ, a sign binary enforces exclusivity. With equal prices simultaneous split slack cancels from settlement; a positive volume penalty removes it. Reported imbalance is computed from physical exchange. Here D_s is executed-path planned wear, I_s import cost and K_s terminal cost (possibly negative continuation value). All battery transitions, initial state, POI, power/SOC and terminal constraints apply on each path. There are *no new auction orders* in this physical-control problem. Forecast economics minimizes

$$\begin{aligned} \min \quad & \sum_s w_s \left[D_s + I_s + K_s - R_s + \kappa_{\mathrm{econ}} \sum_t (b^+_{s,t} + b^-_{s,t}) \right] + \lambda_{\mathrm{risk}} \left[\eta + \frac{1}{1-\alpha} \sum_s w_s \xi_s \right], \\ & \xi_s \geq L_s - \eta, \quad \xi_s \geq 0. \end{aligned} \tag{105}$$

η is unrestricted, because monetary loss can be negative. This differs from nonnegative absolute-volume CVaR in the auction model. Risk excludes the artificial imbalance regularizer, terminal salvage and uncertain cash on already submitted unpriced auctions. Research economic cases default to $\kappa_{\mathrm{econ}} = 0$, independently of the 500 EUR/MWh price-blind penalty; the risk-neutral baseline has $\lambda_{\mathrm{risk}} = 0$.

For unequal directional prices, sign exclusivity prevents artificial positive and negative settlement at once. Equal prices reduce cash to $\hat{\lambda}_{s,t}(b^+_{s,t} - b^-_{s,t})$; slack in the split cannot create cash. Without a positive absolute-volume coefficient, report reconstructed $|\Delta_t g_{s,t} - Q^{\mathrm{sub}}_t|$, not an arbitrary slack split. Nonnegative import costs use an epigraph tight at an optimum with positive coefficient. Only the first control executes; physical protection and rejected-plan fallbacks are logged. Artificial risk, tracking and terminal terms never enter cash profit.

22.2 Observable-history recourse and the value of dependence

Ordinary deterministic/ensemble MPC shares all horizon controls and energy across paths. Adaptive MPC shares the first command but allows future controls to differ only after distinguishable observed PV histories. With lag $\ell \geq 1$ and quantization width q, a branch key is

$$H_{s,t} = \left(n(s,t-1), \lfloor V_{s,0:t-\ell}/q \rfloor\right), \qquad H_{s,t} = H_{r,t} \Rightarrow n(s,t) = n(r,t). \tag{106}$$

Zero quantization uses exact observations; configured nonbranch times retain the previous owner. The previous node prevents recombining different battery histories. Future prices and member labels do not enter the observation key. The prepared longer study uses hourly branching (four

quarters), one-quarter observation lag and 0.1 MW bins. Exact small-ensemble prefixes can identify a member early, so finite-ensemble recourse is still an approximation to an operational information process.

Raw member trajectories preserve their supplied serial structure. Independent per-lead shuffling permutes the *joint* PV/price sample at each lead:

$$(V'_{s,t}, \lambda'_{s,t}) = (V_{\pi_t(s),t}, \lambda_{\pi_t(s),t}). \tag{107}$$

Equal weights and within-lead joint samples are retained while serial dependence is disrupted. Common open-loop controls with an additive risk-neutral objective are invariant to this operation: their feasible POI support and each lead's expected objective are unchanged. Adaptive recourse and pathwise monetary risk provide mechanisms by which dependence can matter; this invariance is a useful negative control, not a claim that dependence never matters for storage.

ECC uses calibrated marginal samples $z_{(1),t} \le \cdots \le z_{(M),t}$ and the raw member's rank $r_{s,t}$:

$$V^{\mathrm{ECC}}_{s,t} = z_{(r_{s,t}),t}. \tag{108}$$

Seeded tie handling is explicit. Price row identities are retained, and this does not estimate a new joint PV–price copula. The construction follows the rank-rearrangement principle of ECC [4]; its marginal calibration must be supplied separately. Calibrated ECC and its shuffled counterpart have the same marginal scores, allowing a dependence contrast under matched control.

The reviewed-target affine bias/spread calibration fits January 1–14 and freezes parameters for January 15–16 and August 2–3. Its two-parameter form in MW is

$$V'_{s,t} = \mathrm{clip}\{\bar{V}_t + aP_{\mathrm{AC}} + c(V_{s,t} - \bar{V}_t), 0, P_{\mathrm{AC}}\}, \tag{109}$$

where normalized bias $a \in [-0.3, 0.3]$ and spread multiplier $c \in [0.25, 3]$ minimize training mean CRPS. The ensemble-mean coefficient is fixed to one; monthly night masks are applied after clipping. On the August two-day transfer, raw versus calibrated CRPS in MW was 0.02554/0.06944 (farm 1), 0.14332/0.15116 (345), 0.19844/0.22464 (901), and 0.17979/0.29941 (910). Calibration worsened all four marginal scores there. Farm 910's 192 execution targets comprise 132 historical-GHI reconstructions and 60 policy night zeros, so that transfer is not independent meter validation.

22.3 Decision-aware scenario reduction

The reduction preserves original complete paths and redistributes probability mass; it does not take a pointwise minimum. Let ϕ_s concatenate scaled features of PV, ramps, cumulative energy deficit/balance, capacity/POI exceedance proxies, price and uncontrolled settlement exposure. Each block is divided by its unit scale and multiplied by the square root of its weight over block length. The implementation greedily approximates

$$\min_{\mathcal{M}:\,|\mathcal{M}|=K} \sum_s w_s \min_{j\in\mathcal{M}} \|\phi_s - \phi_j\|_2, \qquad \tilde{w}_j = \sum_{s:\,j=\mathrm{assign}(s)} w_s. \tag{110}$$

Representatives are source members and retain positive mass, including tie handling for duplicate paths. Features use only current forecasts, known commitments and asset parameters. The distance is a heuristic relevance metric, not a bound on lost MILP value, tail risk or physical feasibility on discarded scenarios. This approximate reduction must not be confused with the exact PV-envelope theorem for the special auction formulation.

22.4 Conditional terminal continuation value

A finite daily horizon can empty or retain storage for reasons unrelated to future opportunities. The research continuation model uses available nominal PV/settlement forecasts and *known* commitments after the MPC horizon. On an energy grid $\mathcal{E}$, define

$$W_t(e) = \max_{e' \in \mathcal{E}\ mfeasible} \{r_t(e, e') + W_{t+1}(e')\}, \qquad W_{T'}(e) = v_F e. \tag{111}$$

For a transition $h = e' - \rho_t e$, $c = [h]_+ / (\eta_c \Delta_t)$ and $d = \eta_d [-h]_+ / \Delta_t$, subject to power limits. Given nominal PV, feasible grid power ranges from $\max(d - c, -P^{\text{imp}})$ to $\min(d - c + V_t, P^{\text{exp}})$. The one-step reward maximizes signed settlement minus import cost and throughput wear; a piecewise-linear maximum can be checked at range endpoints, zero grid flow and the commitment kink. Idle with interpolated continuation is also considered when decay leaves the discrete grid.

The resulting curve is linearly interpolated and enters MPC as $K_s = -W_T(e_{s,T})$ using an exact PWL graph for that supplied curve. The curve itself is approximate: nominal uncertainty, discretized energy, throughput wear without Xu memory, and no unmodeled future auction recourse. It is neither a learned lifetime value nor realized income. A continuation gain must be compared on matched terminal settings and final inventory.

22.5 Information/access and physical-flexibility experiments

The information/access factorial crosses latest versus frozen-initial information $I \in \{0, 1\}$ with DA-only versus all-auction access $A \in \{0, 1\}$. For matched realized profits J_{IA}, the interaction is

$$\Gamma = J_{11} - J_{10} - J_{01} + J_{00}. \tag{112}$$

A positive value indicates complementarity on those cases and dates; it is not identified if cells differ in assets, targets or forecast families. Paired reports require all four cells and preserve their independent state. Separate contrasts double energy capacity, halve power or halve POI capacity; crossing each with marginal calibration tests whether the calibration effect depends on flexibility. For asset setting b, the difference-in-differences is $(J_{\text{cal},b} - J_{\text{raw},b}) - (J_{\text{cal},0} - J_{\text{raw},0})$. These are configured asset counterfactuals, not measured farm upgrades.

22.6 Oracle certification, time budgets and numerical recovery

The main-tree oracle described earlier shares a ten-second budget and requires its stage-one certificate. The MPC sibling additionally supports separate ten-second stage budgets and a declared 1% time-limit fallback. Let U be independently reconstructed feasible absolute imbalance and L the lower bound. The fallback requires $(U - L)/U \leq 0.01$ for positive U, with explicit near-zero handling; the default target solver gap remains 10^{-4}. Refinement is bounded by

$$A \leq U(1 + r) + b. \tag{113}$$

With an approximate incumbent this is an incumbent cap, not a certified $A_{\min}$ cap. Possible excess over the true minimum is at most $U - L + rU + b$, subject to the recorded numerical tolerances. Invalid refinement retains the validated stage-one action; no valid action triggers tracking fallback. Those are different events and are counted separately.

Optional numerical retry tightens feasibility tolerance and sets numerical focus within the remaining economic-stage time budget. It cannot change physical constraints or weaken the imbalance-cap check. The completed 104-job matrix predates these two added retry cases: current configuration has 28 cases, whereas the completed matrix has 26. The current two-window/four-farm plan therefore describes 224 jobs, not 224 completed results. Timing reports separate model construction, native solver runtime, validation and end-to-end replay.

23 Performed trading experiments and numerical evidence

23.1 Independent-day participation and continuous monthly auction studies

The completed January 2025 farm-910 noon-boundary participation run contains 15 integrated profiles on 31 independently initialized days and a sequential all-auction reference. Its all-auction integrated profile ranks first with mean EUR 3,520.06/day, total EUR 109,121.72, daily population standard deviation EUR 1,187.52, mean absolute imbalance 1.69984 MWh/day and mean ending SOC 28.25%. These are price-oracle operating results, not forecast-price earnings or battery-only value. The next-ranked IDA1+IDA2+IDA3 profile averages EUR 2,893.38/day. Independent daily initialization and fee-state reset prevent interpreting their sum as one continuous asset trajectory.

The later continuous participation study instead carries state separately for each of 15 profiles plus sequential and perfect-PV references. Four farms and two forecast variants have saved full-January comparisons. Every strategy has its own SOC, degradation memory, commitments and fee state; the best profile is an ex-post ranking on January. Appendix A gives all 68 matched profile rows. For farm 1, all-auction deterministic/ensemble profit is EUR 10,216.33/10,197.54, versus sequential EUR 1,848.22/1,798.72 and perfect-PV EUR 10,582.72. The large integrated/sequential gap includes perfect-price cross-auction spread opportunities; it is not purely the value of additional PV forecast information.

The separate mixed-model full-month comparisons below use January 1–31 and August 2–31, 2026. Each month starts independently at its configured state; August does not carry January SOC or simulate February–July. The absent July 31 ensemble cycle excludes August 1. Farm 345 uses the declared pvlib override and the other farms retained TCNs. These earlier target/model policies differ from the new reviewed-GHI registry and from the continuous-participation run. The table consistently uses the named `mixed_models` comparison family rather than mixing its revisions with older all-TCN summaries.

Farm/month	Forecast	Profit EUR	Abs. dev. MWh	Wear EUR	Curt. MWh	End SOC %
1/01	Det.	10,319.57	0.665	452.26	0.267	44.14
1/01	51 paths	10,054.81	0.226	531.35	1.835	50.17
1/01	Perfect PV	10,582.72	0.000	224.23	0.000	50.00
1/08	Det.	28,373.64	3.621	900.23	2.219	48.27
1/08	51 paths	26,986.34	0.816	883.62	11.013	49.32
1/08	Perfect PV	28,991.90	0.000	768.58	0.000	50.00
345/01	Det.	20,134.68	48.063	689.45	0.006	19.36
345/01	51 paths	19,643.28	18.672	922.97	2.959	16.67
345/01	Perfect PV	20,956.88	0.000	283.99	0.000	50.00
345/08	Det.	50,554.56	45.594	922.19	0.310	7.58
345/08	51 paths	49,053.88	10.347	1,428.87	14.229	7.07
345/08	Perfect PV	52,615.69	0.000	967.83	0.000	50.00
901/01	Det.	53,451.80	14.022	1,920.99	3.963	93.00
901/01	51 paths	49,170.14	2.071	1,831.25	42.087	93.00
901/01	Perfect PV	56,170.13	0.000	533.91	0.000	50.00
901/08	Det.	177,589.29	38.691	2,646.46	7.991	52.14
901/08	51 paths	159,278.29	3.251	2,627.54	128.051	53.31
901/08	Perfect PV	181,957.15	0.000	1,935.66	0.000	50.00
910/01	Det.	77,704.74	27.099	3,957.66	14.125	48.68
910/01	51 paths	74,492.63	10.151	3,983.35	45.726	49.32
910/01	Perfect PV	83,258.82	0.000	1,295.10	0.000	50.00
910/08	Det.	222,503.76	422.351	3,454.07	0.414	12.35
910/08	51 paths	216,492.01	189.140	5,407.93	52.514	7.66
910/08	Perfect PV	239,605.95	0.000	4,839.14	0.000	50.00

In these saved cases the exact 51-path policy reduces absolute imbalance but also reduces cash

profit relative to deterministic planning and often increases curtailment. This agrees with the common-control exact-delivery envelope's conservative feasibility mechanism; it does not establish that probabilistic forecasting intrinsically reduces value. The constraints, execution rule, calibration and objective determine how uncertainty is used. Farm 910's large August imbalance belongs to the older target/model setting, not to the later fully reconstructed two-day MPC target. Final SOC differences are shown because cash profit contains no terminal salvage valuation.

The envelope equivalence experiment compares 37 matched auction states across August 2–11, farm 1, then two independent ten-day replays. Both saved replays have profit EUR 10,366.671392, absolute imbalance 0.0123333 MWh, wear EUR 283.344853 and final SOC 74.086594%, agreeing to numerical precision. The recovered-curtailment diagnostic differs because it is relative to the retained plan representation. This does not contradict equal actual flows. The fixed-state equivalence follows the stated theorem; equality of independent replay schedules is empirical and can change under alternative optima.

The 100/500 EUR/MWh stochastic-imbalance sensitivities, ensemble-mean versus full paths, selected 10/20-day participation contrasts and PV-only battery counterfactuals have separate comparison artifacts. They are distinct objectives/samples and are not pooled with the exact-delivery monthly table. Their numerical records are preserved in the sensitivity appendix; no artificial planning penalty is relabelled as realized settlement.

23.2 Original current-PV-known two-day MPC diagnostic

The farm-1 January 15–16 TCN study disclosed current-interval realized PV to MPC but kept future PV forecast-only. On a frozen common ledger, tracking profit was EUR 538.25 with 0.217316 MWh absolute imbalance. Deterministic MPC earned EUR 550.53 with 0.236596 MWh; ensemble MPC EUR 554.73 with 0.258667 MWh. Their price-oracle variants earned EUR 541.99 and EUR 541.41. All MPC cases ended at 50% SOC versus tracking 50.677%. In the continuous deterministic case, profit was EUR 556.16 and imbalance 0.361619 MWh; the EUR 17.91 gain included EUR 24.30 more auction revenue from later changed submissions. The 960 recorded MPC decisions had no tracking fallback or timeout. These figures are a software/development diagnostic with retrospective information, not the forecast-current-PV reviewed-target matrix below.

23.3 Reviewed-target October week

For October 1–7, 2025, deterministic penalty MPC versus its own continuous tracking baseline increased cash profit by EUR 44.53, 675.75, 1,157.14 and 1,408.45 for farms 1, 345, 901 and 910. Portfolio absolute imbalance increased 89.91 MWh. The saved independent audit reconstructs 2,688 MPC decisions with no physical/accounting violations, tracking fallback or timeout. It uses a 10^{-6} reconstruction tolerance; this is numerical consistency evidence, not forecast or economic validation. Ending battery energy differs, and MPC has a soft inventory objective absent from tracking. The result therefore does not isolate control intelligence from inventory preference and information.

23.4 Completed reviewed-target January matrix

The January 15–16, 2026 matrix completed 104/104 jobs: 26 cases by four farms, with a separate tracking baseline for each case. Each target has 192 MPC decisions, hence 19,968 target MPC decisions in total. Saved resolved settings use 50% initial SOC, a 50% soft terminal target at 50 EUR/MWh, hourly recourse branching, one-quarter observation lag and 0.1 MW PV bins. The reduced case retains eight paths with seed 17; the monetary-risk case uses $\alpha = 0.9$, $\lambda_{\text{risk}} = 0.5$. Continuation replaces the soft target with a six-hour, 51-energy-point nominal value curve and zero final salvage. Auction solves have 120 seconds and MPC stages 10 seconds each. The frozen

`january_case_settings.json` retains all 26 farm-1 resolved configurations and their source hashes; farm-specific asset parameters remain in the run plan. The following values sum both days and all four farms; they are *not* mean daily gains. Positive imbalance/wear means more absolute deviation/more wear than that case's baseline.

Case	Profit EUR	Abs. dev. MWh	Wear EUR	End energy MWh
adaptive_continuation	331.45	27.795	297.96	-0.122
adaptive_ecc	302.05	24.400	256.79	-0.122
adaptive_monetary_risk	94.83	21.076	284.63	-0.122
adaptive_raw	301.73	23.550	256.51	-0.122
adaptive_reduced	-130.31	32.188	564.56	-0.122
adaptive_shuffled	332.26	26.487	345.75	-0.122
calibrated_double_energy	497.05	30.093	289.66	-0.122
calibrated_half_poi	321.95	24.657	255.43	-0.122
calibrated_half_power	290.14	15.504	118.23	-0.122
calibrated_shuffled	355.97	26.983	342.83	-0.122
causal_auctions	300.53	21.665	303.46	-0.122
causal_fundamentals	249.38	27.548	542.42	-0.122
da_only_frozen	282.72	20.025	110.42	-0.505
da_only_latest	260.57	22.941	220.69	-0.505
deterministic_economic	98.10	12.294	-103.12	-0.122
deterministic_oracle_gap_fallback	218.42	5.573	16.64	0.363
deterministic_penalty	123.66	5.707	17.82	0.337
deterministic_price_oracle	218.42	5.573	16.64	0.363
double_battery_energy	478.19	29.771	296.91	-0.122
ensemble_economic	177.81	20.784	187.46	-0.122
ensemble_oracle_gap_fallback	8.58	10.565	307.30	-0.122
ensemble_penalty	74.81	9.114	186.09	-0.122
ensemble_price_oracle	8.58	10.565	307.30	-0.122
frozen_information	233.48	21.544	155.66	-0.098
half_battery_power	294.71	15.385	121.86	-0.122
half_poi_capacity	309.85	23.568	245.70	-0.122

The deterministic penalty, ensemble penalty and adaptive-raw gains correspond to EUR 61.83, 37.40 and 150.86 per portfolio day. Their absolute-imbalance increases are 2.85, 4.56 and 11.78 MWh per day. Adaptive reduction gives a negative aggregate cash difference of EUR 130.31, whereas raw adaptive gives a positive EUR 301.73. That is evidence that this particular heuristic/sample can alter economic decisions substantially, not a general reduction theorem. Calibrated ECC and raw adaptive have nearly equal total cash differences here, while shuffling and continuation can change them. The whole two-day sample is too small to rank these policies reliably or estimate tail risk.

Some rows change objectives, asset ratings, forecast information or auction access, so subtraction between arbitrary rows does not isolate a single mechanism. For a pure recourse contrast use matched common/adaptive economic objectives and scenarios; for dependence use matched marginals; for flexibility use its declared factorial contrasts. Each baseline belongs to its own case. Per-farm differences, rather than only portfolio totals, appear in Appendix C and the evidence CSV retains absolute target and baseline metrics, auction cash, settlement, charge/discharge and curtailment.

Using absolute target profits (not differences from changing tracking baselines), the four-cell information-by-auction-access interaction is EUR 206.84 summed over the portfolio and two days. The calibration-by-flexibility interactions are EUR 18.54 for doubled battery energy, EUR −4.89 for halved power and EUR 11.77 for halved POI capacity. These are descriptive contrasts with common targets and dates; two calendar days cannot support a reliable confidence interval or a claim about seasonal interactions.

The saved all-job audit reports zero physical/accounting/artifact violations. No target used a

tracking fallback or solver timeout. Eighteen economic oracle refinements across the ordinary and fallback cases were rejected by validation; validated stage-one actions were retained. These are not 18 invalid accepted actions. Accepted cap residuals stayed within the documented allowance. The older raw-target 32-job batch had 18 excessive cap residuals in two oracle jobs and remains superseded evidence; corrected eight-job oracle results have their own audit. Numerical correctness is stated for each audited batch, not inferred for every historical output from current software tests.

23.5 Reviewed-target August farm-910 comparison

On August 2–3, ensemble penalty MPC has cash gain EUR 96.812, extra absolute imbalance 7.542783 MWh and wear reduction EUR 163.559 versus tracking. Adaptive calibrated ECC gains EUR 271.803, with extra imbalance 26.960265 MWh, extra wear EUR 165.250 and four physical safety overrides. Ensemble price oracle loses EUR 62.935, with extra imbalance 7.501351 MWh, wear reduction EUR 119.006 and 40 time-limited decisions. The 1% fallback variant has essentially the same realized outcome. Each has 192 MPC decisions and no tracking fallback; the saved audit reports no physical/accounting violations. All 192 target intervals are reconstructed daylight or policy night zeros. This is a counterfactual control comparison, not measured plant-performance evidence.

23.6 What remains unfinished

The main-tree May all-farm study is explicitly interrupted with no completed comparison mode. Its successful preflight is not a controller result. The prepared annual deterministic study and full January/August ensemble studies still require complete data and final model integration. Reviewed PV inputs end in August; the missing July 31 ensemble cycle remains material for August 1. The current 28-case research plan adds numerical-retry cases beyond the completed 26-case matrix. Four-cycle neural PV training, recent-window rolling price evaluation and bidirectional-neighbor price search are partial at capture. Mayer-style marginal post-processing has implemented formulations but no completed numerical output in the inspected trees. A future run must not use this document's snapshot label as evidence of present process status.

24 Interpretation and current scientific focus

The completed studies support a working chronological decision framework and several useful empirical observations: price-history/fundamental models can beat persistence on specific samples; calibrated marginals and online weights do not improve every season or gate; strict ensemble delivery can lower imbalance while sacrificing cash profit; and physical MPC can change both settlement and subsequent auction decisions. They do not yet establish a deployable annual strategy, an investment return, or a universal architecture ranking.

The current work focuses on forecast-vintage-aligned neural PV models, selection-fixed native-resolution price comparisons, coherent scenario dependence under adaptive control, and the interaction of information with auction access and storage/POI flexibility. Terminal continuation, monetary risk and reduction remain controlled extensions. A calibrated marginal CDF, an exact finite-scenario solve and a numerically reconciled replay answer different questions; none alone establishes real-world reliability.

Statistical units are calendar blocks, shared across farms where prices/weather are common. The four farms are not independent replications of the market. Two-day confidence intervals or extreme-tail estimates are not informative; block bootstrap intervals on larger samples remain conditional on selection, coverage and target construction. Observed/synthetic target performance, native

product regimes, forecast issue availability, optimization gaps, fallback rates and final inventory must accompany any future claim of improvement.

A Complete continuous participation results

These are four independent January 2026 comparisons. D/E denote updating deterministic/frozen ensemble plus updating deterministic forecasts. Profit is EUR, absolute imbalance is MWh and ending SOC is percent. All rows cover January 1–31; each profile carries its own state. Different studies elsewhere in this report must not be merged by farm/date alone.

A.1 Farm 1

Profile	Profit D	Profit E	Abs. D	Abs. E	SOC D	SOC E
DA	1,648.99	1,748.59	9.651	2.301	7.00	36.56
IDA1	1,665.96	1,730.08	10.026	1.852	34.19	38.71
IDA2	1,779.95	1,700.89	7.800	1.638	50.00	52.06
IDA3	1,643.95	1,553.62	5.190	1.137	33.72	43.43
DA+IDA1	4,974.58	5,017.71	9.186	1.656	7.00	38.88
DA+IDA2	6,193.08	6,135.73	7.064	1.505	38.56	52.28
DA+IDA3	5,101.96	5,170.96	6.531	2.130	33.57	43.76
IDA1+IDA2	5,613.30	5,576.99	7.469	1.532	38.66	52.50
IDA1+IDA3	5,018.40	5,136.20	6.317	1.466	41.68	43.65
IDA2+IDA3	4,829.37	4,772.28	5.762	1.562	33.72	43.41
DA+IDA1+IDA2	7,644.73	7,596.64	7.636	1.506	39.64	52.28
DA+IDA1+IDA3	7,585.70	7,685.52	6.321	1.465	44.35	43.60
DA+IDA2+IDA3	8,455.24	8,446.08	5.817	1.565	33.43	43.49
IDA1+IDA2+IDA3	7,967.28	7,963.91	5.805	1.534	33.50	43.49
DA+IDA1+IDA2+IDA3	10,216.33	10,197.54	5.818	1.577	33.50	43.49
Sequential	1,848.22	1,798.72	6.430	1.896	33.38	43.49
Perfect PV	10,582.72	10,582.72	0.000	0.000	50.00	50.00

A.2 Farm 345

Profile	Profit D	Profit E	Abs. D	Abs. E	SOC D	SOC E
DA	6,494.09	4,796.74	3.812	1.148	59.96	62.23
IDA1	6,624.83	4,793.22	4.203	0.882	73.72	92.04
IDA2	6,773.60	4,858.31	3.427	0.735	67.79	91.81
IDA3	5,490.12	3,957.99	1.136	0.095	46.57	49.05
DA+IDA1	11,835.88	10,095.03	3.817	0.732	55.20	75.39
DA+IDA2	13,710.46	11,920.12	3.323	0.660	67.83	92.04
DA+IDA3	12,101.25	10,462.54	2.985	1.029	75.92	89.22
IDA1+IDA2	12,844.82	11,056.18	3.789	0.685	67.86	92.04
IDA1+IDA3	12,140.08	10,382.20	2.549	0.845	75.72	89.26
IDA2+IDA3	11,691.12	9,894.06	2.405	0.836	76.00	89.22
DA+IDA1+IDA2	16,126.41	14,285.36	3.661	0.689	67.82	92.04
DA+IDA1+IDA3	16,201.23	14,471.14	2.622	0.891	75.72	89.26
DA+IDA2+IDA3	17,477.33	15,727.12	2.448	0.832	75.99	89.23
IDA1+IDA2+IDA3	16,701.51	14,941.97	2.553	0.830	76.00	89.26
DA+IDA1+IDA2+IDA3	20,306.16	18,555.04	2.447	0.780	76.00	89.22
Sequential	6,916.97	5,180.87	2.676	0.837	75.91	89.22
Perfect PV	20,956.88	20,956.88	0.000	0.000	50.00	50.00

A.3 Farm 901

Profile	Profit D	Profit E	Abs. D	Abs. E	SOC D	SOC E
DA	16,467.77	12,232.49	48.292	3.262	7.00	47.53
IDA1	16,963.86	11,909.29	47.777	4.615	30.72	93.00
IDA2	17,212.29	12,156.39	33.781	5.321	61.64	93.00
IDA3	12,874.63	9,399.95	16.635	1.258	36.79	76.04
DA+IDA1	31,169.82	26,579.68	45.542	3.173	7.00	93.00
DA+IDA2	36,224.25	31,415.90	32.108	2.686	52.59	92.91
DA+IDA3	31,108.49	27,580.21	30.688	4.949	41.40	92.91
IDA1+IDA2	33,700.14	28,934.16	32.595	3.631	48.18	92.91
IDA1+IDA3	31,246.00	27,144.25	26.845	4.852	35.82	92.91
IDA2+IDA3	30,165.18	25,803.31	23.804	5.321	36.85	90.55
DA+IDA1+IDA2	42,720.38	37,787.34	32.665	3.132	52.59	92.91
DA+IDA1+IDA3	42,455.20	38,338.63	25.408	4.669	41.40	92.91
DA+IDA2+IDA3	45,935.23	41,734.28	23.345	4.315	36.85	89.78
IDA1+IDA2+IDA3	43,795.80	39,536.97	23.229	4.339	36.85	89.78
DA+IDA1+IDA2+IDA3	53,708.73	49,397.35	23.099	4.320	36.85	89.78
Sequential	16,904.75	12,761.53	25.179	5.509	35.12	89.78
Perfect PV	56,170.13	56,170.13	0.000	0.000	50.00	50.00

A.4 Farm 910

Profile	Profit D	Profit E	Abs. D	Abs. E	SOC D	SOC E
DA	19,804.62	16,479.18	85.420	27.383	13.26	7.00
IDA1	20,307.97	16,539.92	80.770	22.410	34.26	38.90
IDA2	20,814.37	17,130.78	79.143	19.284	50.00	61.69
IDA3	16,166.34	13,807.45	49.643	11.613	37.53	54.32
DA+IDA1	42,888.93	39,331.58	77.013	19.185	34.11	39.36
DA+IDA2	50,929.20	47,401.79	73.608	16.857	24.56	59.83
DA+IDA3	43,657.78	40,796.42	75.612	30.074	37.81	53.97
IDA1+IDA2	47,142.12	43,704.32	75.478	18.861	22.51	61.24
IDA1+IDA3	43,370.40	40,379.00	69.401	21.415	38.24	53.79
IDA2+IDA3	41,507.26	38,501.15	71.408	22.878	37.31	54.28
DA+IDA1+IDA2	61,319.41	57,815.61	76.593	18.861	28.59	60.53
DA+IDA1+IDA3	61,252.43	58,336.44	69.447	20.960	37.74	54.13
DA+IDA2+IDA3	66,905.52	63,917.51	70.746	20.507	37.69	54.32
IDA1+IDA2+IDA3	63,345.33	60,500.17	70.554	21.361	37.64	54.32
DA+IDA1+IDA2+IDA3	78,971.91	76,171.83	70.412	21.643	37.64	54.32
Sequential	20,500.95	17,612.81	75.749	24.531	37.51	54.32
Perfect PV	83,258.82	83,258.82	0.000	0.000	50.00	50.00

B Earlier auction sensitivities and battery counterfactuals

These are separate historical comparisons, not revisions to the matched monthly table. Farm 1, January 2026, used the earlier physical/scenario setting. The scalar penalty is a planning deterrent; the table reports actual cash after signed settlement and wear, without deducting that illustrative penalty.

Formulation	PV information	Profit EUR	Abs. MWh	Wear EUR
Exact delivery	Forecast ensemble	9,856.79	1.370	369.46
Exact delivery	Perfect PV	10,582.72	0.000	224.23
Penalty 500 EUR/MWh	Forecast ensemble	10,167.58	4.729	457.24
Penalty 500 EUR/MWh	Perfect PV	10,582.72	0.000	224.23
Penalty 100 EUR/MWh	Forecast ensemble	7,043.63	643.228	114.21
Penalty 100 EUR/MWh	Perfect PV	7,112.42	642.110	90.22

At the lower penalty the model takes very large short positions: forecast-case auction receipts EUR 97,335.24 are offset by settlement EUR $-90,177.39$. This is a strong objective-sensitivity diagnostic, not evidence that a 100 EUR/MWh coefficient represents actual imbalance economics. Even perfect PV does not prevent deliberate planned deviations in that formulation.

In the separately saved January battery-minus-PV-only comparison, incremental cash profit was EUR 57.77 (deterministic), 162.22 (51 paths), and 83.67 (perfect PV). Added wear was EUR 520.36, 369.46 and 224.23; absolute imbalance changed by -5.041, -1.884 and effectively zero MWh. These are operational cash differences under the saved initial/terminal conditions, not annual battery returns or capital-cost benefits.

The accompanying `auction_sensitivities.csv` retains 220 rows from these comparisons and the farm-specific 10/20-day four-policy participation studies, including ensemble-mean control, PV-only references, ending energy and saved validation-warning counts where provided. Periods and objectives must be matched before comparing rows. Sources added during the final audit carry their own addition timestamp in the evidence ledger; the ongoing-search status snapshot remains frozen at the original capture time.

C January MPC results by farm and case

Every number is target controller minus its own tracking baseline over the two days January 15–16, 2026. End MWh is final stored-energy difference. These are cash comparisons without inventory valuation. The CSV also preserves absolute target/baseline totals and solver diagnostics.

Farm / case	Profit EUR	Abs. MWh	Wear EUR	End MWh
1 / adaptive_continuation	28.60	2.621	35.83	0.000
1 / adaptive_ecc	12.91	2.428	32.22	0.000
1 / adaptive_monetary_risk	-0.34	1.840	29.93	0.000
1 / adaptive_raw	18.40	2.389	31.49	0.000
1 / adaptive_reduced	14.57	1.730	19.06	0.000
1 / adaptive_shuffled	14.34	2.154	27.05	0.000
1 / calibrated_double_energy	37.64	2.631	21.67	0.000
1 / calibrated_half_poi	14.14	2.281	28.96	0.000
1 / calibrated_half_power	20.10	1.337	10.35	0.000
1 / calibrated_shuffled	18.63	2.404	32.15	0.000
1 / causal_auctions	27.72	2.164	38.36	0.000
1 / causal_fundamentals	30.81	2.424	52.14	0.000
1 / da_only_frozen	19.34	1.799	16.09	0.000
1 / da_only_latest	10.17	2.218	24.87	0.000
1 / deterministic_economic	2.52	1.059	-8.27	0.000
1 / deterministic_oracle_gap_fallback	27.51	0.387	-0.58	0.000
1 / deterministic_penalty	25.48	0.412	-0.08	0.000
1 / deterministic_price_oracle	27.51	0.387	-0.58	0.000
1 / double_battery_energy	50.79	2.941	27.47	0.000
1 / ensemble_economic	13.24	2.273	29.31	0.000
1 / ensemble_oracle_gap_fallback	17.66	0.530	14.07	0.000
1 / ensemble_penalty	17.93	0.471	9.91	0.000

Farm / case	Profit EUR	Abs. MWh	Wear EUR	End MWh
1 / ensemble_price_oracle	17.66	0.530	14.07	0.000
1 / frozen_information	18.50	1.824	23.38	0.000
1 / half_battery_power	26.68	1.492	13.24	0.000
1 / half_poi_capacity	26.73	2.377	29.60	0.000
345 / adaptive_continuation	45.32	4.378	-9.13	-0.122
345 / adaptive_ecc	34.12	3.927	-11.78	-0.122
345 / adaptive_monetary_risk	24.50	3.518	-16.57	-0.122
345 / adaptive_raw	34.32	3.871	-13.77	-0.122
345 / adaptive_reduced	46.06	3.863	-12.05	-0.122
345 / adaptive_shuffled	54.27	4.204	-5.49	-0.122
345 / calibrated_double_energy	53.79	4.347	-11.17	-0.122
345 / calibrated_half_poi	34.12	3.927	-11.78	-0.122
345 / calibrated_half_power	50.26	2.748	-30.57	-0.122
345 / calibrated_shuffled	54.27	4.204	-5.49	-0.122
345 / causal_auctions	82.58	4.097	2.10	-0.122
345 / causal_fundamentals	82.64	4.660	27.31	-0.122
345 / da_only_frozen	121.76	3.363	-7.10	-0.505
345 / da_only_latest	117.86	3.910	5.49	-0.505
345 / deterministic_economic	21.86	2.365	-55.54	-0.122
345 / deterministic_oracle_gap_fallback	21.39	1.293	-27.49	-0.028
345 / deterministic_penalty	16.40	1.199	-27.49	-0.028
345 / deterministic_price_oracle	21.39	1.293	-27.49	-0.028
345 / double_battery_energy	53.61	4.347	-11.17	-0.122
345 / ensemble_economic	30.46	3.333	-23.91	-0.122
345 / ensemble_oracle_gap_fallback	22.17	1.295	-14.87	-0.122
345 / ensemble_penalty	9.79	1.142	-22.41	-0.122
345 / ensemble_price_oracle	22.17	1.295	-14.87	-0.122
345 / frozen_information	47.78	3.386	-19.88	-0.098
345 / half_battery_power	50.17	2.748	-30.57	-0.122
345 / half_poi_capacity	34.39	3.837	-14.41	-0.122
901 / adaptive_continuation	74.79	5.413	62.53	0.000
901 / adaptive_ecc	70.93	5.051	64.15	0.000
901 / adaptive_monetary_risk	8.45	4.700	81.96	0.000
901 / adaptive_raw	76.12	4.733	58.32	0.000
901 / adaptive_reduced	-50.15	6.956	120.50	0.000
901 / adaptive_shuffled	80.62	6.135	98.63	0.000
901 / calibrated_double_energy	119.15	6.586	73.97	0.000
901 / calibrated_half_poi	64.59	5.030	66.06	0.000
901 / calibrated_half_power	72.77	3.419	29.90	0.000
901 / calibrated_shuffled	78.74	5.717	86.96	0.000
901 / causal_auctions	77.48	4.564	61.47	0.000
901 / causal_fundamentals	29.66	6.093	117.38	0.000
901 / da_only_frozen	46.09	4.008	26.41	0.000
901 / da_only_latest	67.54	4.538	33.84	0.000
901 / deterministic_economic	19.22	2.426	-14.61	0.000
901 / deterministic_oracle_gap_fallback	5.22	1.502	32.07	0.365
901 / deterministic_penalty	-7.48	1.653	32.07	0.365
901 / deterministic_price_oracle	5.22	1.502	32.07	0.365
901 / double_battery_energy	91.73	6.579	75.43	0.000
901 / ensemble_economic	36.38	4.236	48.64	0.000
901 / ensemble_oracle_gap_fallback	-140.68	3.270	142.36	0.000
901 / ensemble_penalty	-68.02	2.818	77.99	0.000
901 / ensemble_price_oracle	-140.68	3.270	142.36	0.000
901 / frozen_information	44.34	4.296	40.17	0.000
901 / half_battery_power	73.06	3.458	30.63	0.000
901 / half_poi_capacity	76.12	4.733	58.32	0.000
910 / adaptive_continuation	182.75	15.383	208.73	0.000
910 / adaptive_ecc	184.10	12.994	172.20	0.000

Farm / case	Profit EUR	Abs. MWh	Wear EUR	End MWh
910 / adaptive_monetary_risk	62.23	11.019	189.31	0.000
910 / adaptive_raw	172.89	12.557	180.47	0.000
910 / adaptive_reduced	-140.79	19.639	437.05	0.000
910 / adaptive_shuffled	183.04	13.995	225.56	0.000
910 / calibrated_double_energy	286.48	16.529	205.19	0.000
910 / calibrated_half_poi	209.10	13.419	172.20	0.000
910 / calibrated_half_power	147.01	8.000	108.56	0.000
910 / calibrated_shuffled	204.34	14.659	229.21	0.000
910 / causal_auctions	112.76	10.840	201.52	0.000
910 / causal_fundamentals	106.27	14.371	345.59	0.000
910 / da_only_frozen	95.52	10.855	75.03	0.000
910 / da_only_latest	65.00	12.275	156.49	0.000
910 / deterministic_economic	54.50	6.444	-24.70	0.000
910 / deterministic_oracle_gap_fallback	164.31	2.391	12.64	0.026
910 / deterministic_penalty	89.27	2.442	13.33	0.000
910 / deterministic_price_oracle	164.31	2.391	12.64	0.026
910 / double_battery_energy	282.05	15.904	205.19	0.000
910 / ensemble_economic	97.73	10.942	133.42	0.000
910 / ensemble_oracle_gap_fallback	109.43	5.469	165.74	0.000
910 / ensemble_penalty	115.11	4.682	120.60	0.000
910 / ensemble_price_oracle	109.43	5.469	165.74	0.000
910 / frozen_information	122.86	12.038	111.99	0.000
910 / half_battery_power	144.80	7.687	108.56	0.000
910 / half_poi_capacity	172.61	12.622	172.20	0.000

D Price experiment inventory

Identifiers below are relative to the MPC tree's `outputs/energy_price_forecasts/`. "Records" counts manifest result/model entries, not independent observations or total weekly refits. Derived comparison reports can have zero such entries while retaining completed group metrics. Status is frozen at capture. Complete does not certify numerical convergence, causal availability or untouched evaluation. Per-run periods and all candidate metrics are in `price_run_inventory.csv` and `price_candidate_results.csv` beside the snapshot.

Saved run identifier	Status	Records
`chronological_2026_v3`	complete	30
`european_context_ablation_v1`	complete	75
`european_context_previous_day_v1`	complete	75
`extended_2024_training_v2`	complete	27
`nord_da_2026_huber_search_v1`	complete	56
`nord_da_2026_previous_ida_prices_v1`	complete	140
`nord_da_2026_previous_idas_ablation_v1`	complete	0
`nord_da_2026_previous_idas_selected_rolling_v1`	complete	10
`nord_da_2026_recent_training_windows_v1`	complete	96
`nord_da_2026_recent_windows_selected_rolling_v1`	running	5
`nord_da_curve_monthly_diagnostics_v1`	complete	0
`nord_da_curves_rolling_v1`	complete	15
`nord_da_daily_curves_search_v1`	complete	315
`nord_da_expanded_context_search_v1`	complete	243
`nord_da_french_balance_matched_v1`	complete	108
`nord_da_per_slot_search_v1`	complete	40
`nord_da_resolution_holdout_v1`	complete	10
`nord_da_resolution_search_v1`	complete	60
`nord_ida_2024_hourly_exports_v1`	complete	243
`nord_ida_2025_selected_rolling_holdout_v1`	complete	27

Saved run identifier	Status	Records
nord_ida_2025_terna_load_ablation_v1	complete	0
nord_ida_2025_terna_load_matched_v1	complete	324
nord_ida_2025_terna_selected_rolling_holdout_v1	complete	21
nord_ida_2026_huber_search_v1	complete	144
nord_ida_bidirectional_neighbors_search_v1	running	470
nord_ida_cleared_context_2025_v1	complete	486
nord_ida_cleared_context_search_v1	complete	486
nord_ida_cleared_monthly_diagnostics_v1	complete	0
nord_ida_cleared_rolling_v1	complete	27
nord_ida_french_balance_ablation_v1	complete	0
nord_ida_french_balance_matched_v1	complete	324
nord_ida_nearest_auction_spreads_v1	complete	324
nord_ida_per_slot_search_v1	complete	72
nord_ida_swiss_border_matched_v1	complete	324
nord_ida_swiss_reservoir_ablation_v1	complete	0
nord_ida_swiss_reservoir_matched_v1	complete	324
nord_ida_temporal_ensemble_v1	complete	0
nord_ida_temporal_hourwise_v1	complete	0
nord_ida_temporal_stacking_v1	complete	0
nord_intraday_settlement_resolution_search_v1	complete	120

E Evidence ledger and review method

The report uses the full main LaTeX as its base, reconciles the sibling feature LaTeX and MPC/price LaTeX, and checks the optimizer, physical helpers, information ownership, replay accounting, forecast adapters and effective configuration. Numerical evidence is read from completed run records, compact CSV exports and saved independent audits; no optimization or training was rerun for this report. Where only an older report survives, that weaker evidence is stated. The source ledger below maps every pinned input to its checkout; full SHA-256 values and captured working-tree state are in `docs/report_evidence/2026-11-04/snapshot.json`.

The frozen CSV tables are a compact report snapshot, not replacement live run manifests. They preserve comparison labels and target provenance. The main working tree had unrelated ongoing PV-training modifications and an existing price-data inventory; those are outside this documentation change. The two sibling working trees were read as evidence rather than modified or merged. The independently written presentation document is `docs/research_overview.tex`; it retains the scientific assumptions without implementation identifiers and places its detailed formulation in appendices.

ID / checkout	Evidence path
S001 / main	`docs/results/pv_reviewed_hpo/historical_ghi_all_farms_night3000_v1/top_three_metrics.csv`
S002 / main	`outputs/continuous_optimization/comparisons/farm_1/2026_01_mixed_models_pv_comparison/monthly_comparison.csv`
S003 / main	`outputs/continuous_optimization/comparisons/farm_1/2026_08_mixed_models_pv_comparison/monthly_comparison.csv`
S004 / main	`outputs/continuous_optimization/comparisons/farm_345/2026_01_mixed_models_pv_comparison/monthly_comparison.csv`
S005 / main	`outputs/continuous_optimization/comparisons/farm_345/2026_08_mixed_models_pv_comparison/monthly_comparison.csv`
S006 / main	`outputs/continuous_optimization/comparisons/farm_901/2026_01_mixed_models_pv_comparison/monthly_comparison.csv`
S007 / main	`outputs/continuous_optimization/comparisons/farm_901/2026_08_mixed_models_pv_comparison/monthly_comparison.csv`

ID / checkout	Evidence path
S008 / main	outputs/continuous_optimization/comparisons/farm_910/2026_01_mixed_models_pv_comparison/monthly_comparison.csv
S009 / main	outputs/continuous_optimization/comparisons/farm_910/2026_08_mixed_models_pv_comparison/monthly_comparison.csv
S010 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_continuous_participation_forecast_comparison/monthly_comparison.csv
S011 / main	outputs/continuous_optimization/comparisons/farm_345/2026_01_continuous_participation_forecast_comparison/monthly_comparison.csv
S012 / main	outputs/continuous_optimization/comparisons/farm_901/2026_01_continuous_participation_forecast_comparison/monthly_comparison.csv
S013 / main	outputs/continuous_optimization/comparisons/farm_910/2026_01_continuous_participation_forecast_comparison/monthly_comparison.csv
S014 / MPC	outputs/energy_research/reviewed_january_matrix_v1/manifest.json
S015 / MPC	outputs/energy_price_forecasts/chronological_2026_v3/manifest.json
S016 / MPC	outputs/energy_price_forecasts/european_context_ablation_v1/manifest.json
S017 / MPC	outputs/energy_price_forecasts/european_context_previous_day_v1/manifest.json
S018 / MPC	outputs/energy_price_forecasts/extended_2024_training_v2/manifest.json
S019 / MPC	outputs/energy_price_forecasts/nord_da_2026_huber_search_v1/manifest.json
S020 / MPC	outputs/energy_price_forecasts/nord_da_2026_previous_ida_prices_v1/manifest.json
S021 / MPC	outputs/energy_price_forecasts/nord_da_2026_previous_idas_ablation_v1/manifest.json
S022 / MPC	outputs/energy_price_forecasts/nord_da_2026_previous_idas_selected_rolling_v1/manifest.json
S023 / MPC	outputs/energy_price_forecasts/nord_da_2026_recent_training_windows_v1/manifest.json
S024 / MPC	outputs/energy_price_forecasts/nord_da_2026_recent_windows_selected_rolling_v1/manifest.json
S025 / MPC	outputs/energy_price_forecasts/nord_da_curve_monthly_diagnostics_v1/manifest.json
S026 / MPC	outputs/energy_price_forecasts/nord_da_curves_rolling_v1/manifest.json
S027 / MPC	outputs/energy_price_forecasts/nord_da_daily_curves_search_v1/manifest.json
S028 / MPC	outputs/energy_price_forecasts/nord_da_expanded_context_search_v1/manifest.json
S029 / MPC	outputs/energy_price_forecasts/nord_da_french_balance_matched_v1/manifest.json
S030 / MPC	outputs/energy_price_forecasts/nord_da_per_slot_search_v1/manifest.json
S031 / MPC	outputs/energy_price_forecasts/nord_da_resolution_holdout_v1/manifest.json
S032 / MPC	outputs/energy_price_forecasts/nord_da_resolution_search_v1/manifest.json
S033 / MPC	outputs/energy_price_forecasts/nord_ida_2024_hourly_exports_v1/manifest.json
S034 / MPC	outputs/energy_price_forecasts/nord_ida_2025_selected_rolling_holdout_v1/manifest.json
S035 / MPC	outputs/energy_price_forecasts/nord_ida_2025_terna_load_ablation_v1/manifest.json
S036 / MPC	outputs/energy_price_forecasts/nord_ida_2025_terna_load_matched_v1/manifest.json
S037 / MPC	outputs/energy_price_forecasts/nord_ida_2025_terna_selected_rolling_holdout_v1/manifest.json

ID / checkout	Evidence path
S038 / MPC	outputs/energy_price_forecasts/nord_ida_2026_huber_search_v1/manifest.json
S039 / MPC	outputs/energy_price_forecasts/nord_ida_bidirectional_neighbors_search_v1/manifest.json
S040 / MPC	outputs/energy_price_forecasts/nord_ida_cleared_context_2025_v1/manifest.json
S041 / MPC	outputs/energy_price_forecasts/nord_ida_cleared_context_search_v1/manifest.json
S042 / MPC	outputs/energy_price_forecasts/nord_ida_cleared_monthly_diagnostics_v1/manifest.json
S043 / MPC	outputs/energy_price_forecasts/nord_ida_cleared_rolling_v1/manifest.json
S044 / MPC	outputs/energy_price_forecasts/nord_ida_french_balance_ablation_v1/manifest.json
S045 / MPC	outputs/energy_price_forecasts/nord_ida_french_balance_matched_v1/manifest.json
S046 / MPC	outputs/energy_price_forecasts/nord_ida_nearest_auction_spreads_v1/manifest.json
S047 / MPC	outputs/energy_price_forecasts/nord_ida_per_slot_search_v1/manifest.json
S048 / MPC	outputs/energy_price_forecasts/nord_ida_swiss_border_matched_v1/manifest.json
S049 / MPC	outputs/energy_price_forecasts/nord_ida_swiss_reservoir_ablation_v1/manifest.json
S050 / MPC	outputs/energy_price_forecasts/nord_ida_swiss_reservoir_matched_v1/manifest.json
S051 / MPC	outputs/energy_price_forecasts/nord_ida_temporal_ensemble_v1/manifest.json
S052 / MPC	outputs/energy_price_forecasts/nord_ida_temporal_hourwise_v1/manifest.json
S053 / MPC	outputs/energy_price_forecasts/nord_ida_temporal_stacking_v1/manifest.json
S054 / MPC	outputs/energy_price_forecasts/nord_intraday_settlement_resolution_search_v1/manifest.json
S055 / features	outputs/price_benchmark/20261023T133005867139Z/results.csv
S056 / main	outputs/pv_reviewed_hpo/four_cycle_neural_night3000_gpu_parallel_v1/status.json
S057 / main	outputs/pv_reviewed_hpo/four_cycle_neural_night3000_gpu_v1/status.json
S058 / main	outputs/pv_reviewed_hpo/four_cycle_neural_night3000_multi_gpu_v1/status.json
S059 / main	outputs/pv_reviewed_hpo/four_cycle_neural_night3000_v1/status.json
S060 / main	outputs/pv_reviewed_hpo/four_cycle_neural_night3000_v2/status.json
S061 / main	outputs/pv_reviewed_hpo/historical_ghi_all_farms_night3000_v1/status.json
S062 / main	outputs/pv_reviewed_hpo/historical_ghi_all_farms_v1/status.json
S063 / main	AGENTS.md
S064 / main	configs/standard.yaml
S065 / main	docs/pv_probabilistic_online_crps_reference.tex
S066 / main	docs/results/pv_online_crps_january_august_2026/metrics.csv
S067 / main	docs/results/probabilistic_pv_january_august_2026/comparison/paired_comparison.csv
S068 / main	docs/results/pv_realized_irradiance_2025_2026/README.md
S069 / main	docs/results/cleaned_pv_hpo/all_farms_cleaned_16_trials/selected_model_summary.csv
S070 / main	docs/results/realtime_mpc_2026_01_15_16/continuous_controller_comparison.csv

ID / checkout	Evidence path
S071 / main	docs/results/realtime_mpc_2026_01_15_16/frozen_commitments_controller_comparison.csv
S072 / main	outputs/realtime_mpc/may_2026_all_farms/MAY_2026_ALL_FARMS_SUMMARY.md
S073 / main	models/pv/reviewed_ghi_2026_selected/manifest.json
S074 / main	outputs/continuous_optimization/comparisons/farm_1/envelope_replay_2026_08_02_to_11/period_comparison.csv
S075 / main	outputs/continuous_optimization/comparisons/farm_1/envelope_replay_2026_08_02_to_11/matched_state_solves.csv
S076 / main	outputs/participation_studies/gme_nord_farm910_2025-01_current_settings_sequential_ida3_noon/json/manifest.json
S077 / main	outputs/participation_studies/gme_nord_farm910_2025-01_current_settings_sequential_ida3_noon/csv/profile_ranking.csv
S078 / features	docs/mathematical_model.tex
S079 / features	docs/price_benchmark.md
S080 / features	outputs/price_features/history_2024_coverage.json
S081 / features	outputs/price_benchmark/20261023T133005867139Z/manifest.json
S082 / features	outputs/price_benchmark/20261023T133151054322Z/results.csv
S083 / features	outputs/price_benchmark/20261023T134106810433Z/results.csv
S084 / features	outputs/price_benchmark/20261023T134324360680Z/results.csv
S085 / MPC	docs/realtime_mpc_price_forecasting.tex
S086 / MPC	docs/energy_market_research_extensions.md
S087 / MPC	docs/reviews/mpc_results_overview.md
S088 / MPC	docs/reviews/price_huber_convergence.md
S089 / MPC	docs/reviews/price_da_2026_previous_idas.md
S090 / MPC	docs/reviews/price_ida_2025_selected_rolling.md
S091 / MPC	docs/reviews/continuous_mpc_reviewed_october_v1.json
S092 / MPC	docs/reviews/continuous_mpc_reviewed_farm910_august_v1.json
S093 / MPC	outputs/energy_research/reviewed_january_matrix_audit.json
S094 / MPC	outputs/research_inputs/pv_calibration_reviewed/calibration_summary.json
S095 / main	src/optimization/model.py
S096 / main	src/optimization/mpc.py
S097 / main	src/optimization/physical.py
S098 / main	src/optimization/degradation.py
S099 / main	src/optimization/pv_envelope.py
S100 / main	src/simulation/replay.py
S101 / main	src/simulation/continuous.py
S102 / main	src/simulation/continuous_inputs.py
S103 / main	src/forecasting/inference/pv_reviewed_hpo.py
S104 / features	src/forecasting/price_benchmark/models.py
S105 / features	src/forecasting/price_benchmark/data.py
S106 / features	src/forecasting/price_benchmark/evaluation.py
S107 / MPC	src/optimization/mpc.py
S108 / MPC	src/optimization/adaptive_mpc.py
S109 / MPC	src/scenario_research.py
S110 / MPC	src/forecasting/continuation.py
S111 / MPC	src/forecasting/energy_prices.py
S112 / MPC	src/forecasting/price_search.py
S113 / MPC	src/forecasting/price_search_models.py
S114 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_participation_days_20_four_policies/period_comparison.csv
S115 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_participation_ten_days_four_policies/period_comparison.csv
S116 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_pv_battery_comparison/monthly_comparison.csv
S117 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_pv_deterministic_vs_scenarios/monthly_comparison.csv

ID / checkout	Evidence path
S118 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_pv_scenarios_comparison/monthly_comparison.csv
S119 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_pv_scenarios_penalty100_comparison/monthly_comparison.csv
S120 / main	outputs/continuous_optimization/comparisons/farm_345/2026_01_participation_ten_days_four_policies/period_comparison.csv
S121 / main	outputs/continuous_optimization/comparisons/farm_901/2026_01_participation_ten_days_four_policies/period_comparison.csv
S122 / main	outputs/continuous_optimization/comparisons/farm_910/2026_01_participation_ten_days_four_policies/period_comparison.csv
S123 / main	outputs/continuous_optimization/comparisons/farm_1/2026_01_pv_battery_comparison/battery_effect_by_policy.csv
S124 / MPC	src/forecasting/pv_scenario_calibration.py

References

[1] B. Xu, J. Zhao, T. Zheng, E. Litvinov and D. S. Kirschen, *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*. Author preprint and published paper links retained by the repository: https://arxiv.org/abs/1707.04567 and https://doi.org/10.1109/TPWRS.2017.2733339.

[2] pvlib Python documentation. The installed library implements the solar-position, single-axis tracking, isotropic transposition, Faiman and PVWatts functions called by `forecasting/models/pv.py`. https://pvlib-python.readthedocs.io/.

[3] Gestore dei Mercati Energetici, Technical Rule 03 rev. 13 MPE, valid from 29 September 2025, and rev. 14 MPE, valid from 13 January 2026. The project uses their gate/result timing with explicit local study settings; DA publication at 13:30 is identified as an assumption, while recorded times are preferred. GME's 29 September 2025 notice describes 15-minute MGP products from delivery date 1 October 2025. https://www.mercatoelettrico.org/Portals/0/Documents/en-US/20250930DTF03rev13MPE_En.pdf https://gme.mercatoelettrico.org/Portals/0/Documents/en-US/20260113DTF03rev14MPE_en.pdf https://www.mercatoelettrico.org/en-us/Home/NoticesandAnnouncements/NoticesandAnnouncementsME?id=7101

[4] R. Schefzik, T. L. Thorarinsdottir and T. Gneiting, *Uncertainty Quantification in Complex Simulation Models Using Ensemble Copula Coupling*, Statistical Science 28(4), 616–640, 2013. https://arxiv.org/abs/1302.7149.

[5] `scoringRules`, normal, censored-normal and truncated-normal score reference. https://search.r-project.org/CRAN/refmans/scoringRules/html/scores_norm.html.

[6] Repository source files listed in Section 1 and the effective configuration saved with each run are authoritative for implemented behavior. Conceptual references do not replace inspection of those inputs and policies.